

CIVL6415, Semester 2, 2024

TRAFFIC ANALYSIS AND SIMULATION

MODULE 4

Macroscopic traffic models

Lecture Week 8

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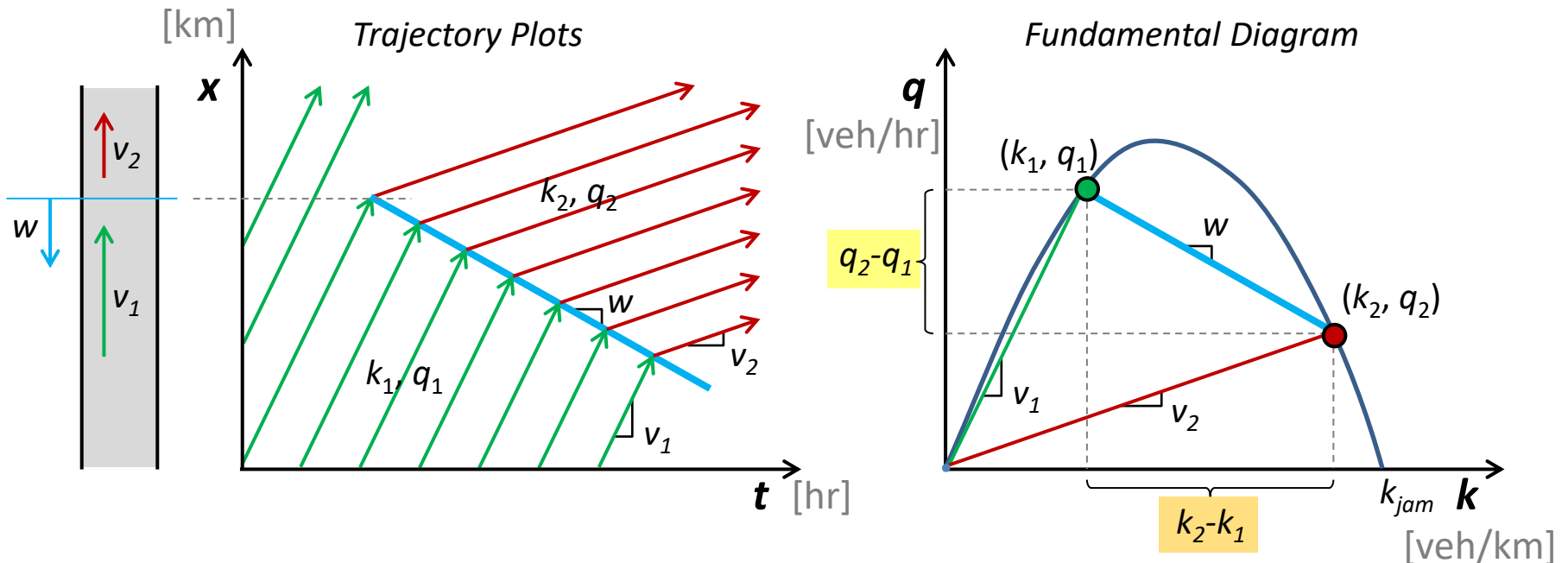
- Arrival / Departure or Input / Output curves
- Shockwaves
- Shockwave velocity equation
- **Types of bottlenecks**
 - Temporary bottlenecks
 - Stationary bottlenecks
 - Moving bottlenecks
- Flow conservation law

Velocity of Shockwaves

Shockwave Velocity (w)

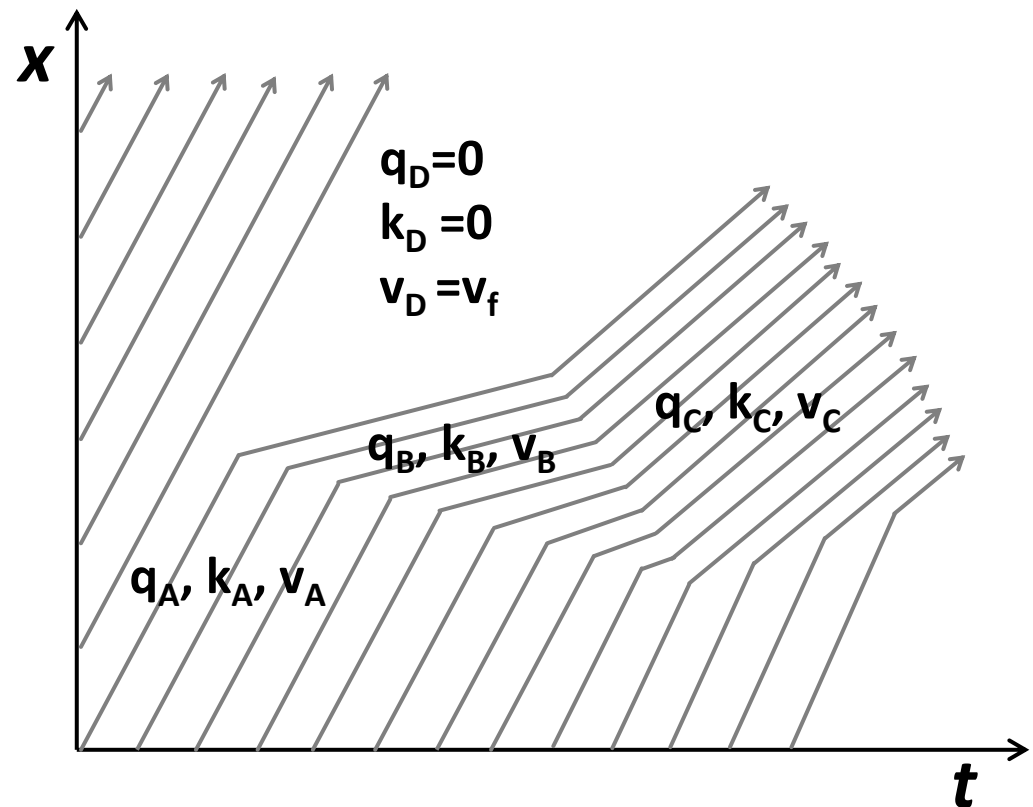
$$w = \frac{q_2 - q_1}{k_2 - k_1} \quad [\text{km/hr}]$$

Geometric Interpretation on Fundamental Diagram



■ Key Parameters of a Shock Wave

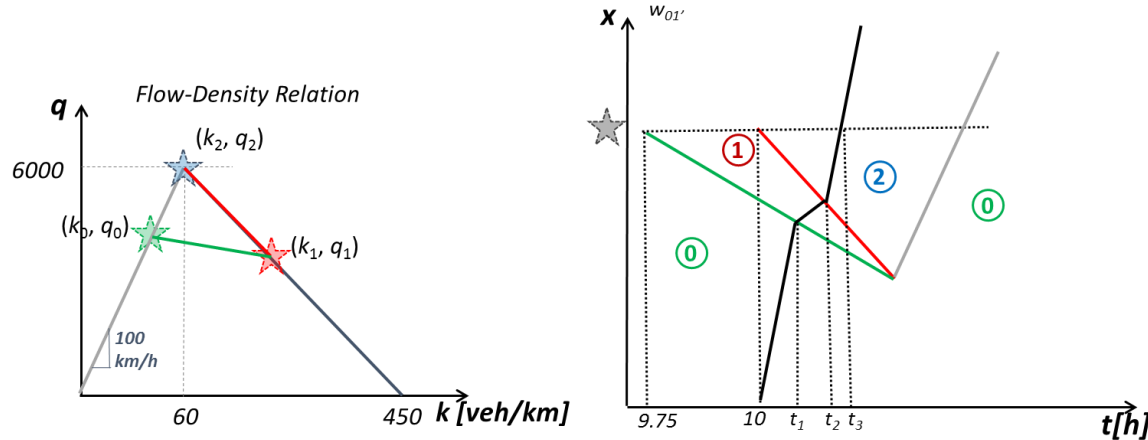
- Identify traffic states
- the point at which it starts,
- the point at which it ends, and
- **the velocity of the shock wave**
(speed + direction).



Temporary Bottlenecks

- Traffic on a three-lane freeway observes a triangular flow-density relationship with free-flow speed $u_f=100\text{km/h}$, capacity $c=2000\text{veh/h/lane}$, jam density $k_j=150\text{veh/km/lane}$. Flow in the freeway from upstream is 5000veh/h , while the ramp flow is negligible (almost zero).
- At 9:45am, an accident occurs at point A. The vehicles involved close one lane of traffic (total capacity reduces to 4000veh/h) for 15min and afterwards the scene is clear (capacity returns to 6000veh/h). Lucy enters the freeway via the ramp at point B (upstream of point A) at 10:00am. Distance AB is equal to 6.5km.
- Space-time diagram with all the shockwaves
- Lucy's trajectory

Exercise



$$q_0 = 5000 \text{ veh/h}, k_0 = 5000/100 = 50 \text{ veh/km}, v_0 = 100 \text{ km/h}$$

$$q_2 = 6000 \text{ veh/h}, k_2 = 6000/100 = 60 \text{ veh/km}, v_2 = 100 \text{ km/h}$$

$$q_1 = 4000 \text{ veh/h}$$

Using similar triangles; $k_1 = 60 + \frac{450-60}{6000} * (6000 - 4000) = 190 \text{ veh/km}$

$$v_1 = \frac{4000}{190} = 21.05 \text{ km/h}$$

$$w_{0-1} = \frac{q_0 - q_1}{k_0 - k_1} = \frac{5000 - 4000}{50 - 190} = -7.14 \text{ km/h}$$

$$w_{1-2} = \frac{q_1 - q_2}{k_1 - k_2} = \frac{4000 - 6000}{190 - 60} = -15.38 \text{ km/h}$$

Question

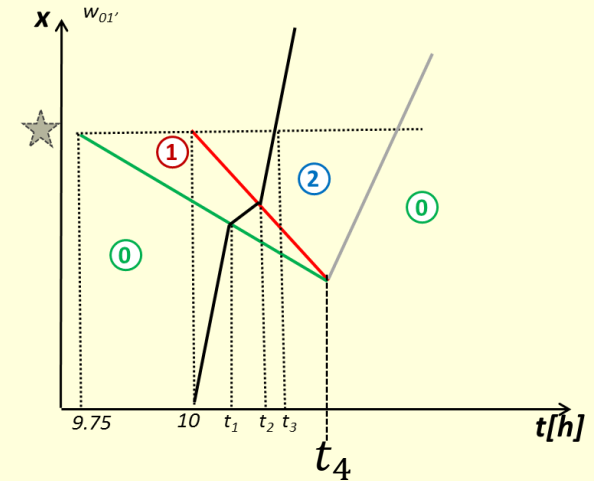
q-k Curves at Lane-Drop Bottleneck

Calculate the time when the congestion ends, t_4 [h].

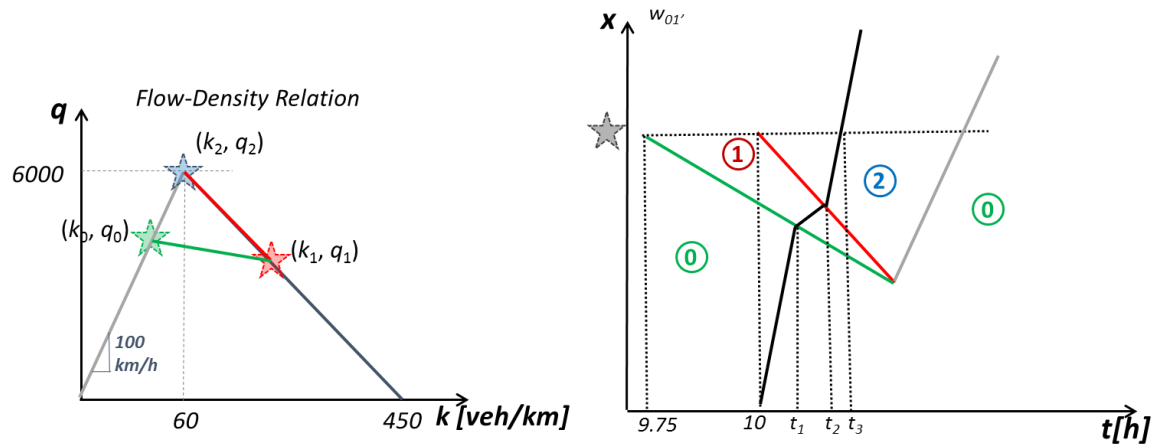
$$w_{0-1} = -7.14 \text{ km/h}$$

$$w_{1-2} = -15.38 \text{ km/h}$$

- a. 10.111
- b. 10.150
- c. 10.166
- d. 10.200
- e. 10.217



Exercise

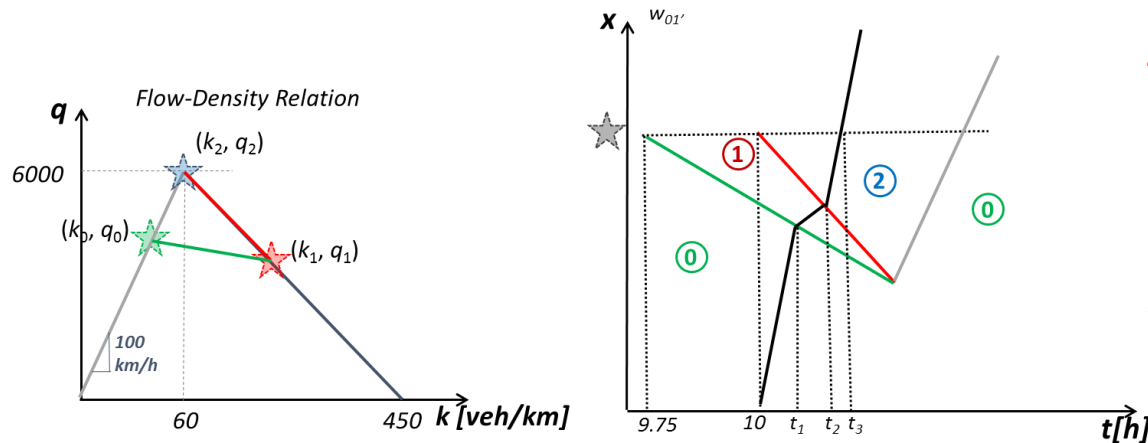


When does Lucy hit the queue? Point t_1 in the space-time diagram.

$$\begin{aligned} v_0 * (t_1 - 10) &= 6.5 + w_{0-1} * (t_1 - 9.75) \\ 100 * (t_1 - 10) &= 6.5 - 7.14 * (t_1 - 9.75) \\ &\rightarrow t_1 = 10.044 \text{ h} \end{aligned}$$

Make sure $t_1 < t_4$.

Exercise



Note that by the time Lucy crosses the accident location, the scene is clear

When does Lucy leave the queue? Point t_2 in the space-time diagram.

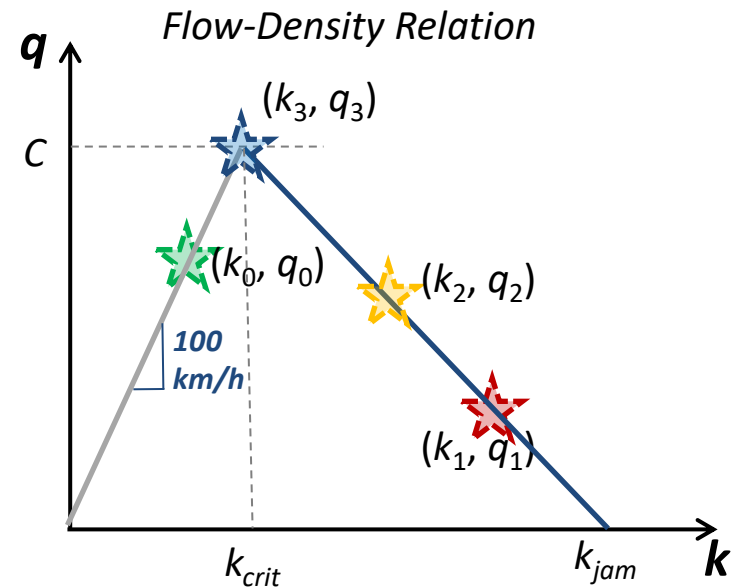
$$\begin{aligned} v_0 * (t_1 - 10) + v_1 * (t_2 - t_1) &= 6.5 + w_{1-2} * (t_2 - 10) \\ 100 * 0.044 + 21.05 * (t_2 - 10.044) &= 6.5 - 15.38 * (t_2 - 10) \\ &\rightarrow t_2 = 10.083 \text{ h} \end{aligned}$$

When does Lucy cross the accident scene? Point t_3 in the space-time diagram.

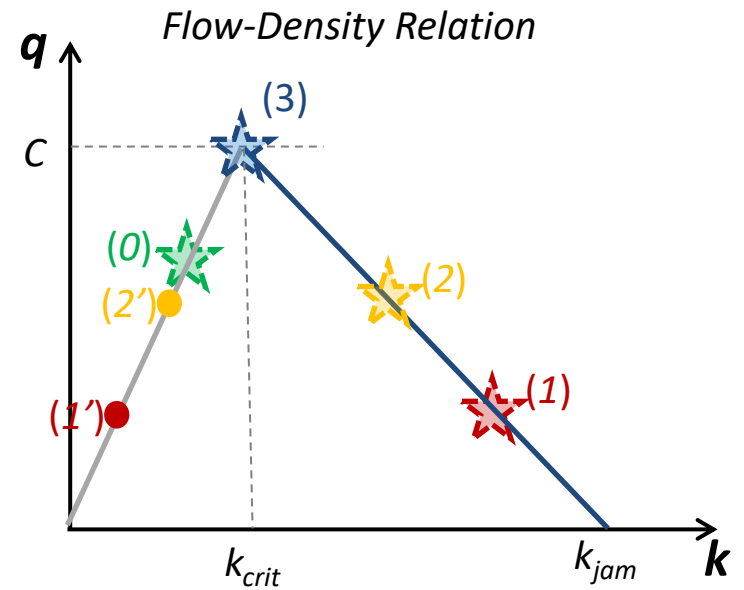
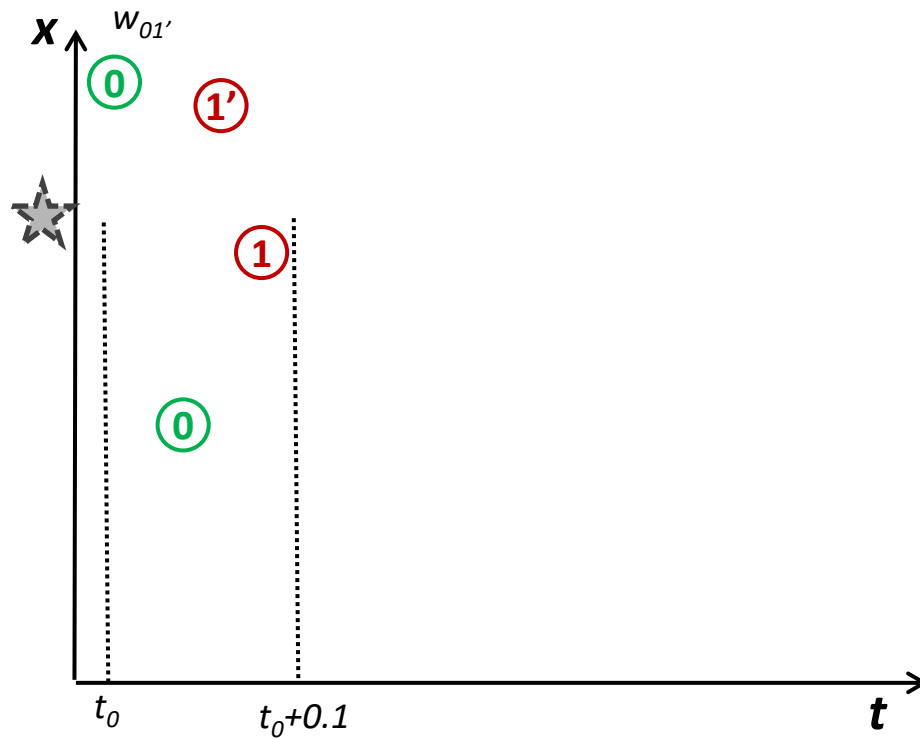
$$\begin{aligned} v_0 * (t_1 - 10) + v_1 * (t_2 - t_1) + v_2 * (t_3 - t_2) &= 6.5 \\ 100 * 0.044 + 21.05 * 0.039 + 100 * (t_3 - 10.083) &= 6.5 \\ &\rightarrow t_3 = 10.096 \text{ h} \end{aligned}$$

Exercise

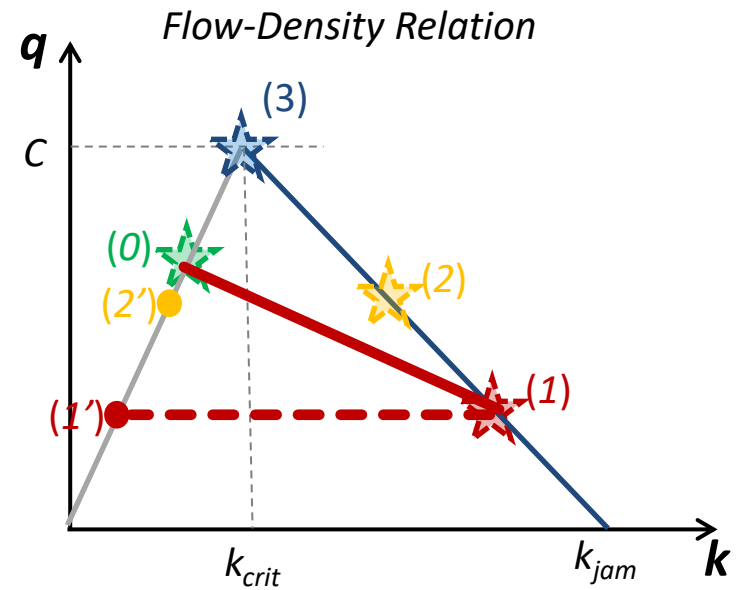
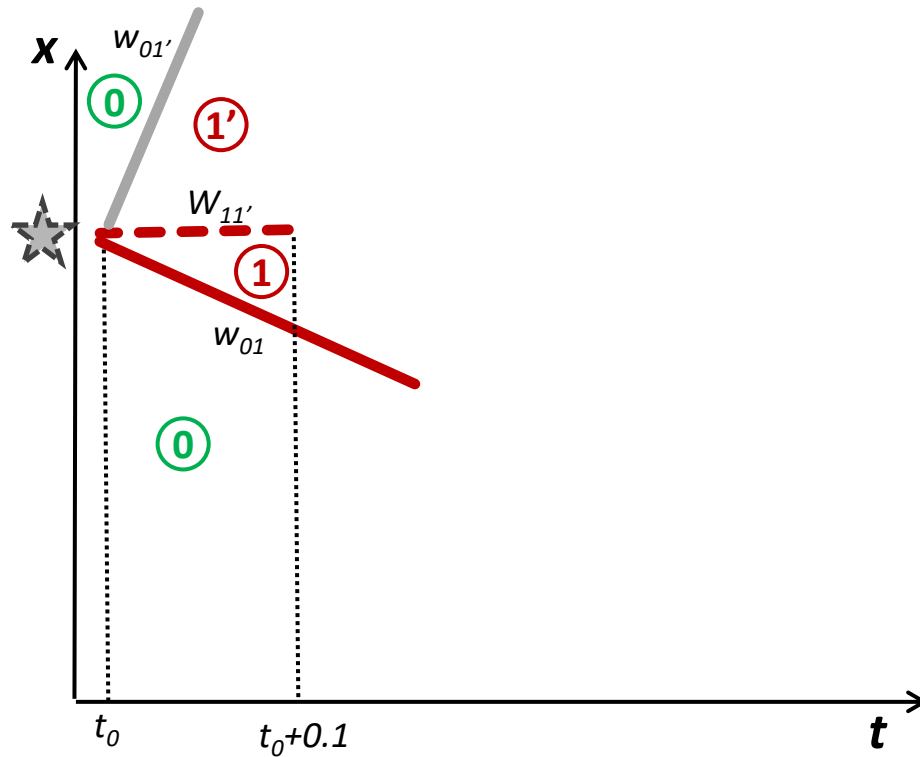
- A freeway with 3 lanes and maximum capacity $C = 6000$ veh/h
- Triangular fundamental diagram, where $v_f = 100$ km/h, $k_{jam} = k_{crit} * 5$
- Demand = $\frac{3}{4} * C$
- Incident at $t_0 = 0$, Response time = 6 min, Clearance time = 42 min
- One lane remains open during response
- Two lanes remain open during clearance



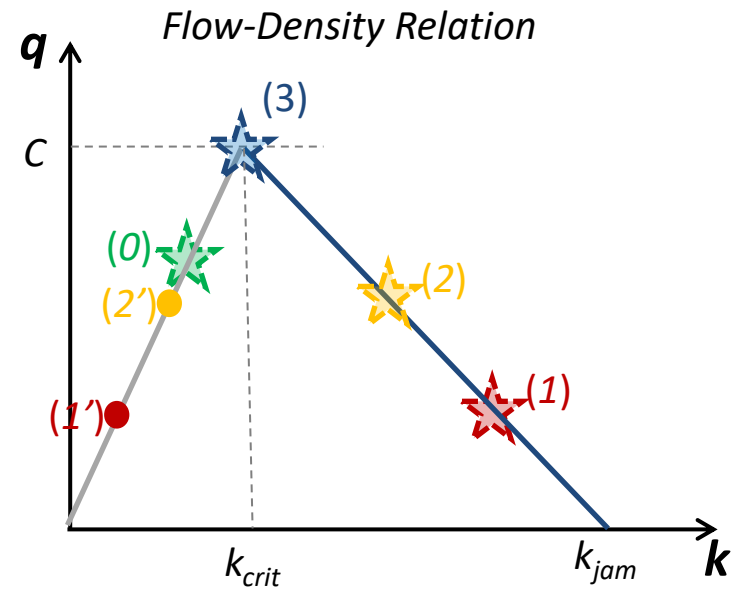
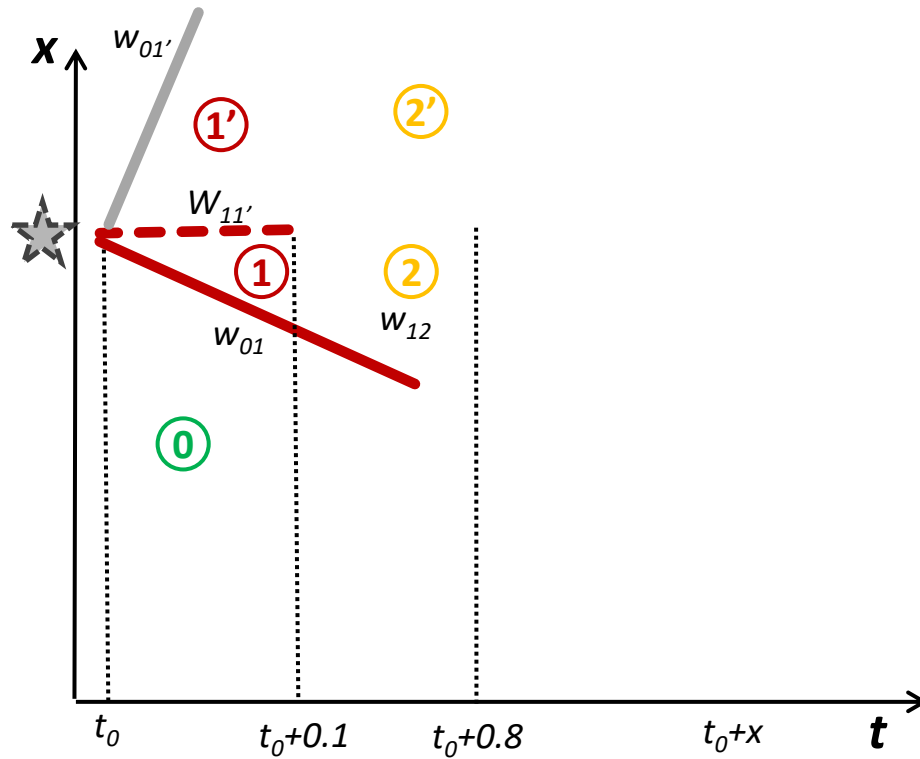
Exercise



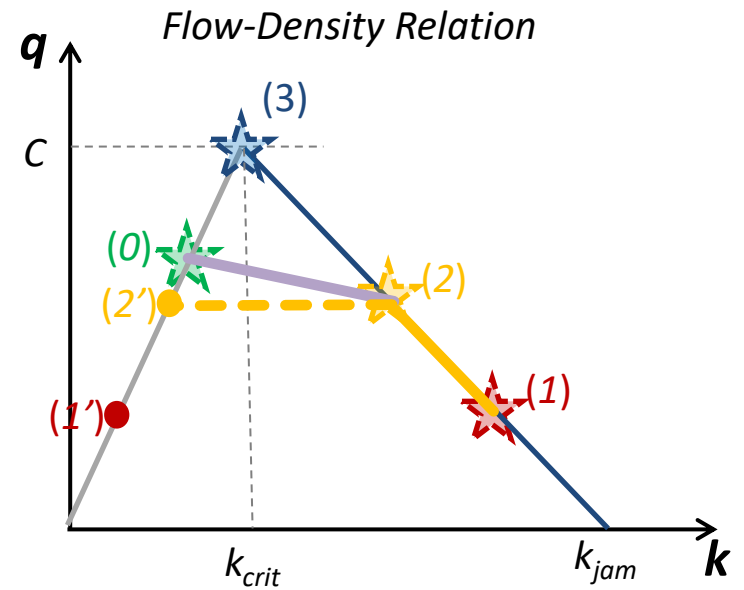
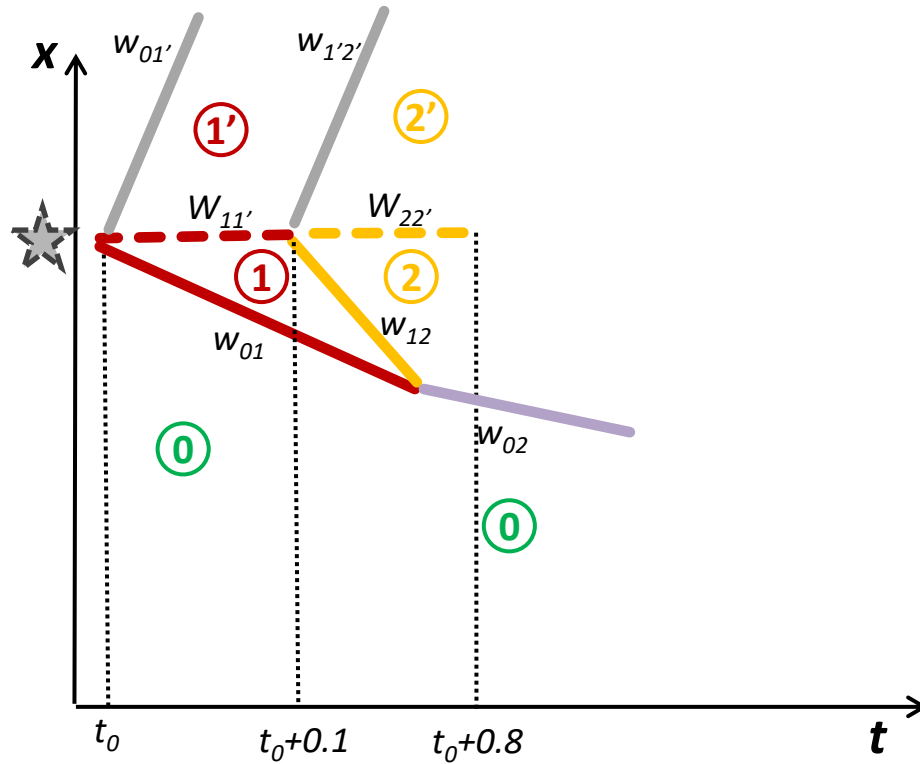
Exercise



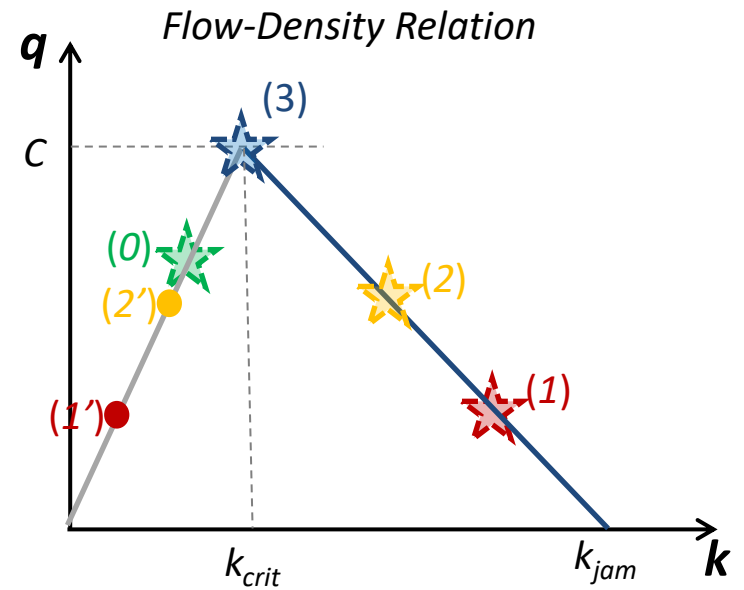
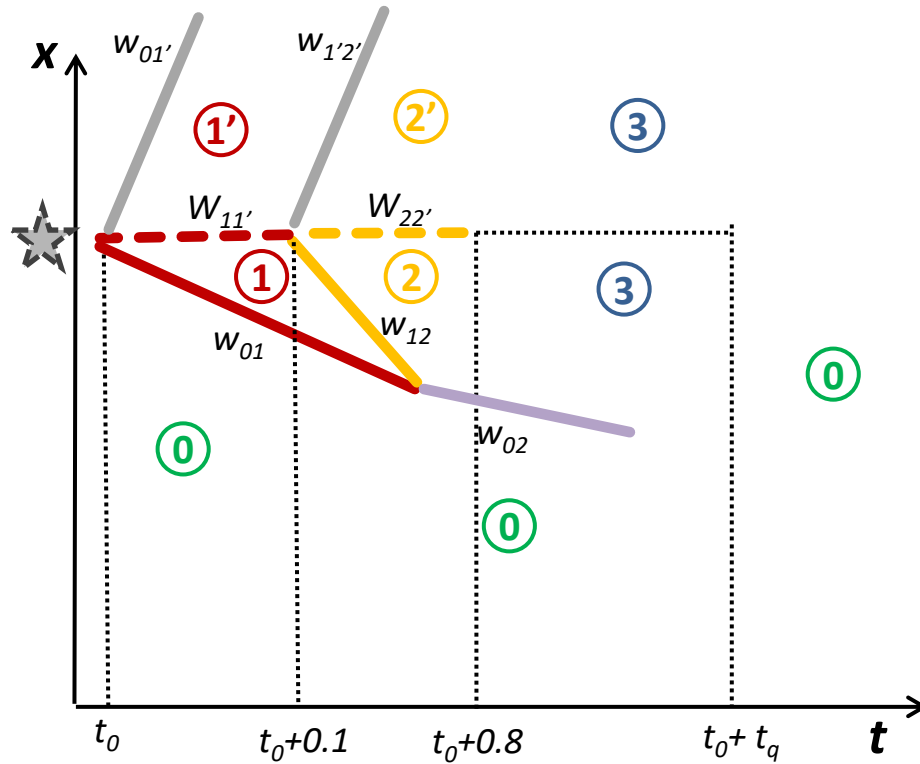
Exercise



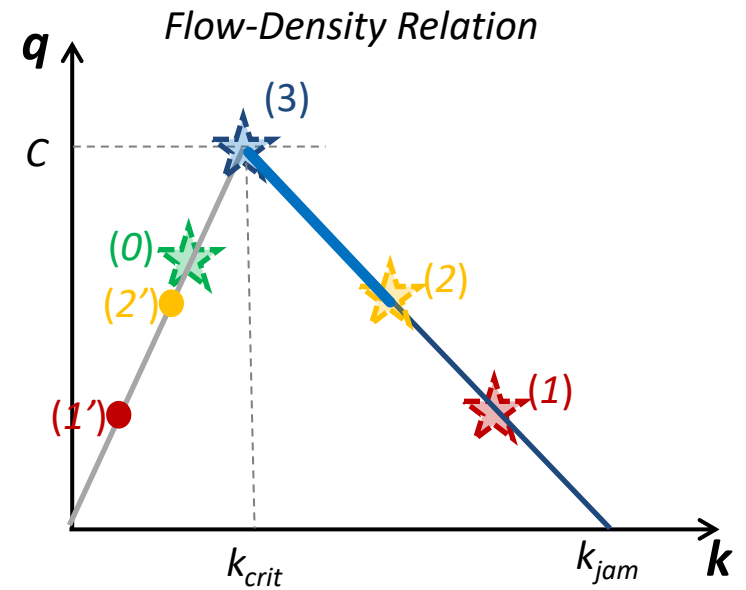
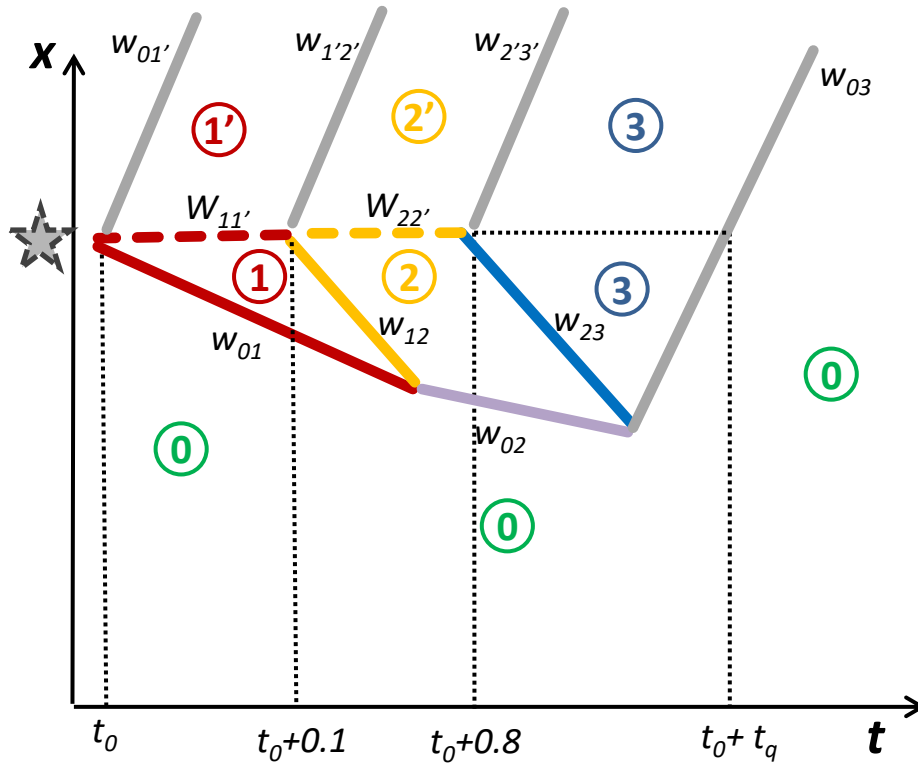
Exercise



Exercise



Exercise



Exercise

3 lanes, $C = 6000$ veh/h, Demand = $\frac{3}{4} * C$

$v_f = 100$ km/h, $k_{jam} = k_{crit} * 5$

$k_{crit} = 6000/100 = 60$ veh/km

$k_{jam} = 300$ veh/km

$q_2 = \frac{2}{3} * C = 4000$ veh/h

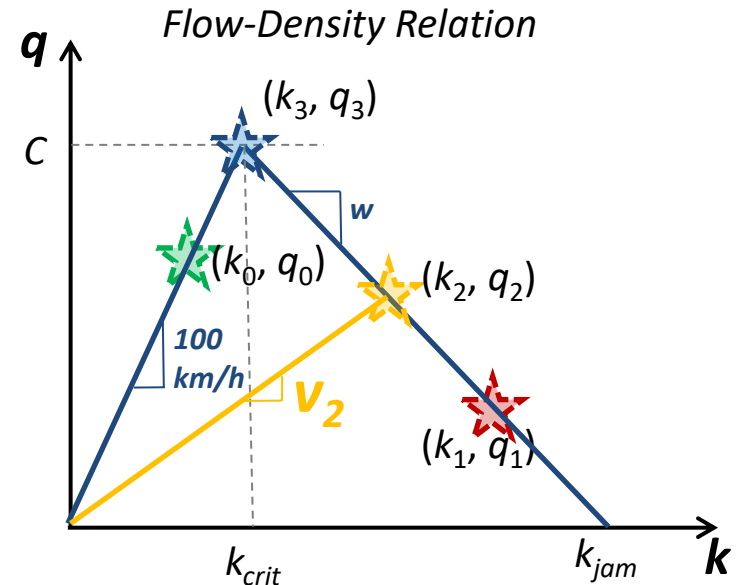
$w = (q_{jam} - q_{crit}) / (k_{jam} - k_{crit}) = -25$ km/h

$w = (q_2 - q_{crit}) / (k_2 - k_{crit})$

$k_2 = k_{crit} + 1/w * (q_2 - q_{crit})$

$k_2 = 60 - 1/25 * (4000 - 6000) = 140$ veh/km

$v_2 = q_2/k_2 = 4000/140 = 28.57$ km/h



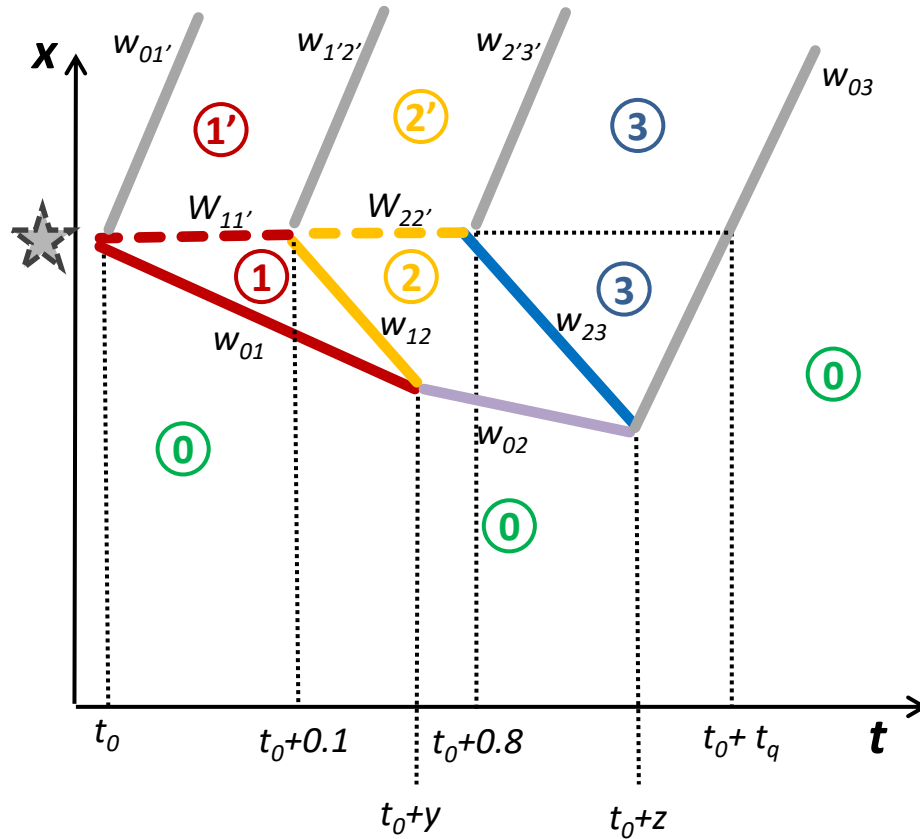
$q_0 = 4500$ veh/h, $k_0 = 45$ veh/km

$q_1 = 2000$ veh/h, $k_1 = 220$ veh/km

$q_2 = 4000$ veh/h, $k_2 = 140$ veh/km

$q_3 = 6000$ veh/h, $k_3 = 300$ veh/km

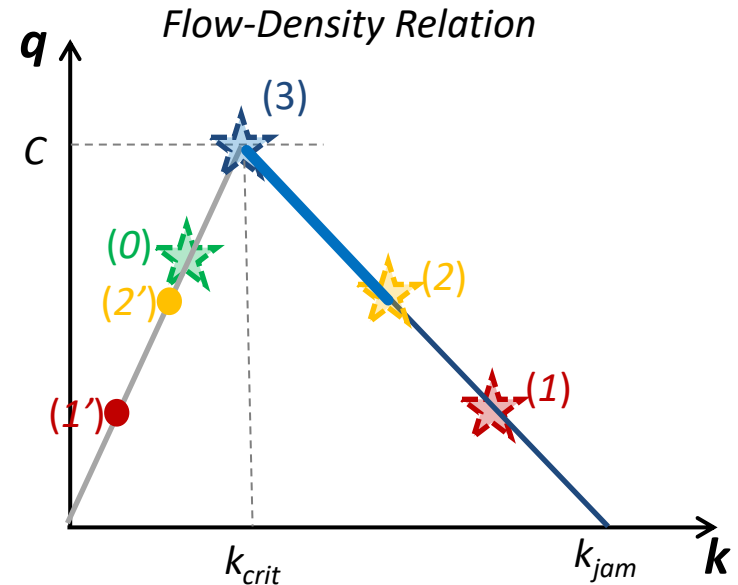
Exercise



$$w_{01} = (2000-4500)/(220-45) = -14.29 \text{ km/h}$$

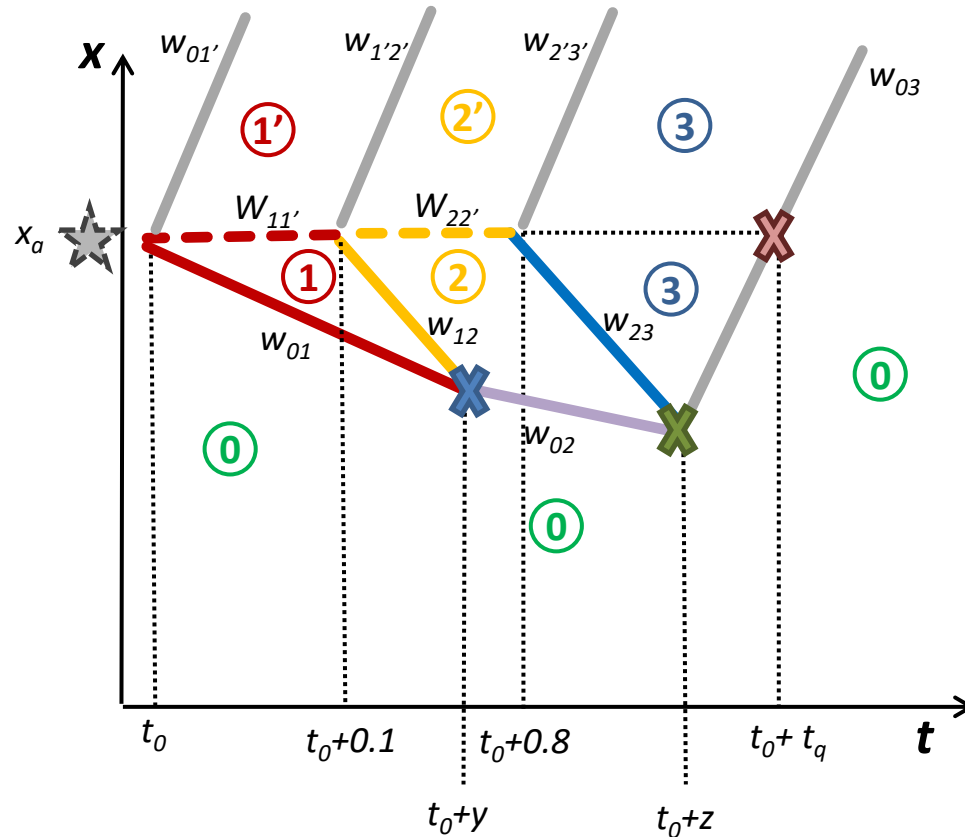
$$w_{02} = (4000-4500)/(140-45) = -5.26 \text{ km/h}$$

$$w_{12} = w_{23} = 6000/240 = -25 \text{ km/h}$$



$$v_f = w_{01'} = w_{1'2'} = w_{2'3'} = w_{03} = 100 \text{ km/h}$$

Exercise



X

$$x_a + w_{01} * y = x_a + w_{12} * (y - 0.1)$$

$$\rightarrow y = 0.23h$$

X

$$x_a + w_{12} * (y - 0.1) + w_{02} * (z - y) = x_a + w_{23} * (z - 0.8)$$

$$\rightarrow z = 1.12h$$

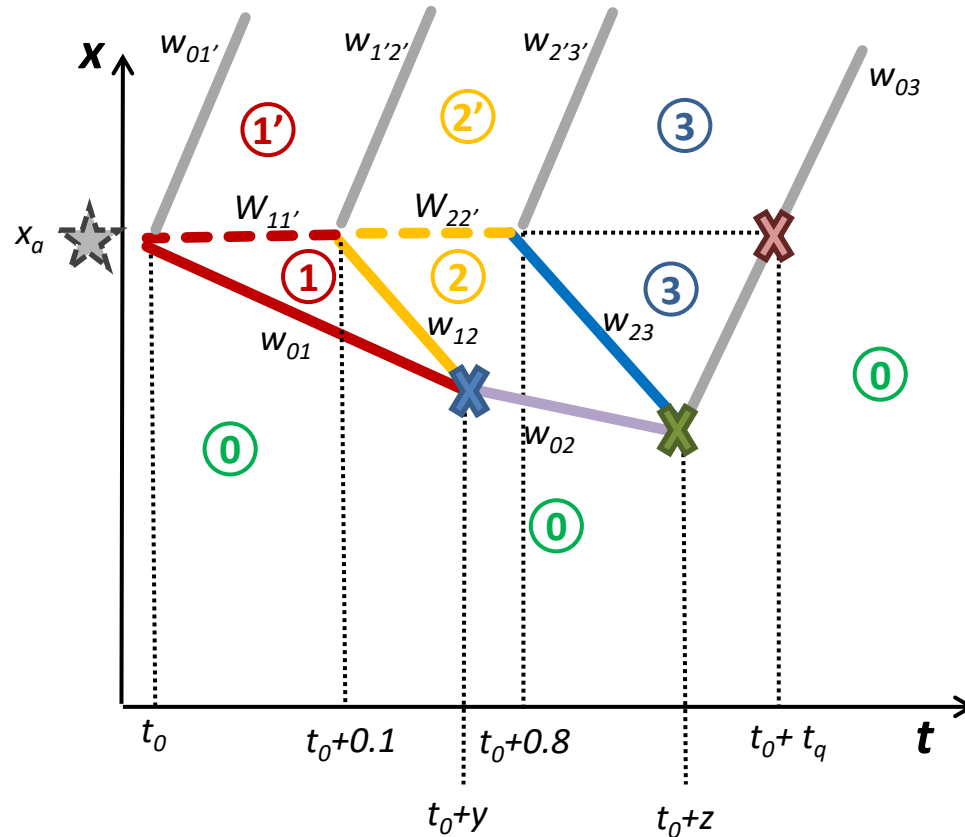
$$w_{01} = -14.29 \text{ km/h}$$

$$w_{02} = -5.26 \text{ km/h}$$

$$w_{12} = w_{23} = -25 \text{ km/h}$$

$$v_f = w_{01'} = w_{1'2'} = w_{2'3'} = w_{03} = 100 \text{ km/h}$$

Exercise



X

$$x_a + w_{23} * (z - 0.8) + v_f * (t_a - z) = x_a$$

$$-25 * (1.12 - 0.8) + 100 * (t_a - 1.12) = 0$$

$$\rightarrow t_a = 1.20h$$

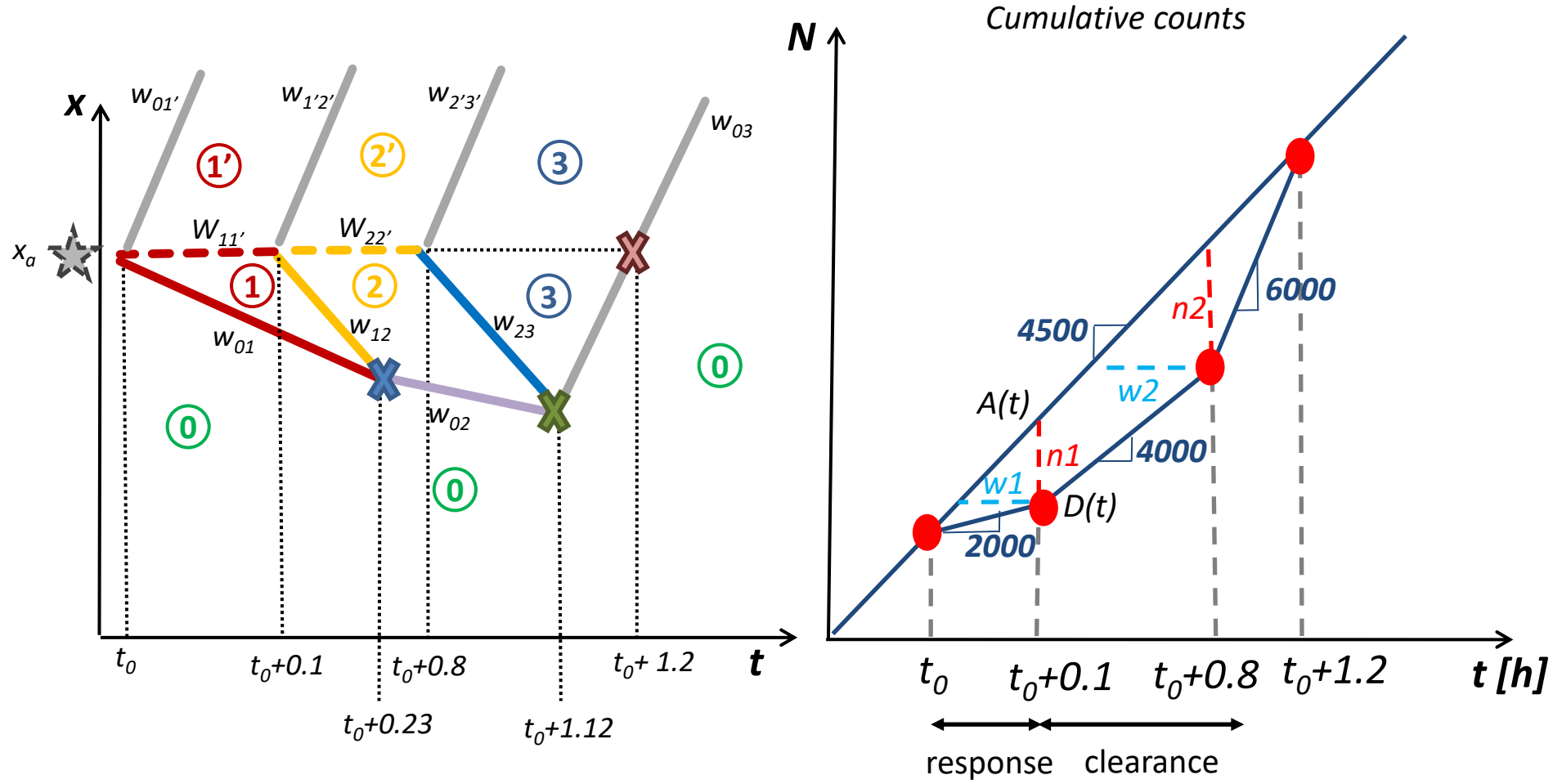
$$w_{01} = -14.29 \text{ km/h}$$

$$w_{02} = -5.26 \text{ km/h}$$

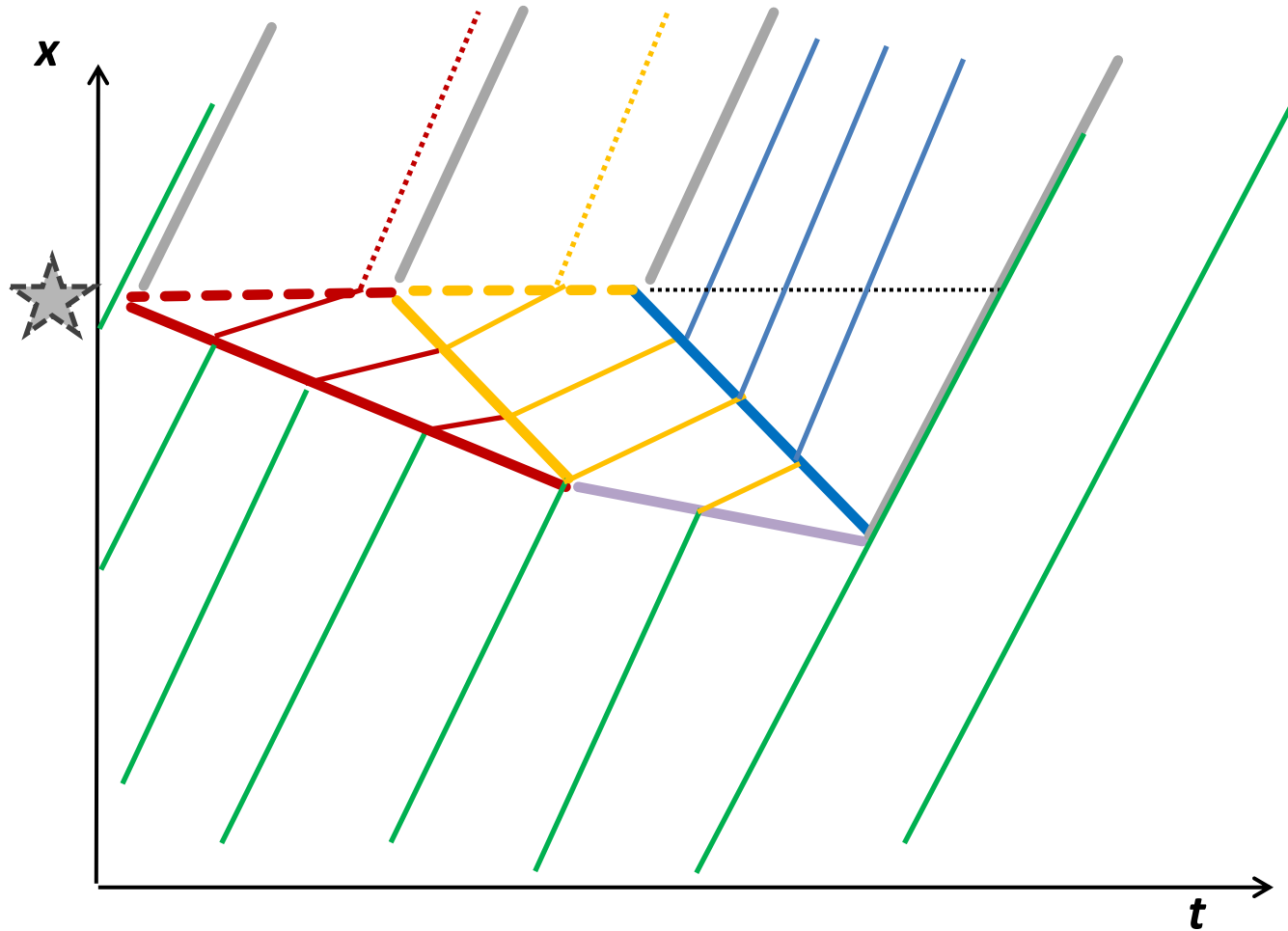
$$w_{12} = w_{23} = -25 \text{ km/h}$$

$$v_f = w_{01'} = w_{1'2'} = w_{2'3'} = w_{03} = 100 \text{ km/h}$$

Exercise



Exercise

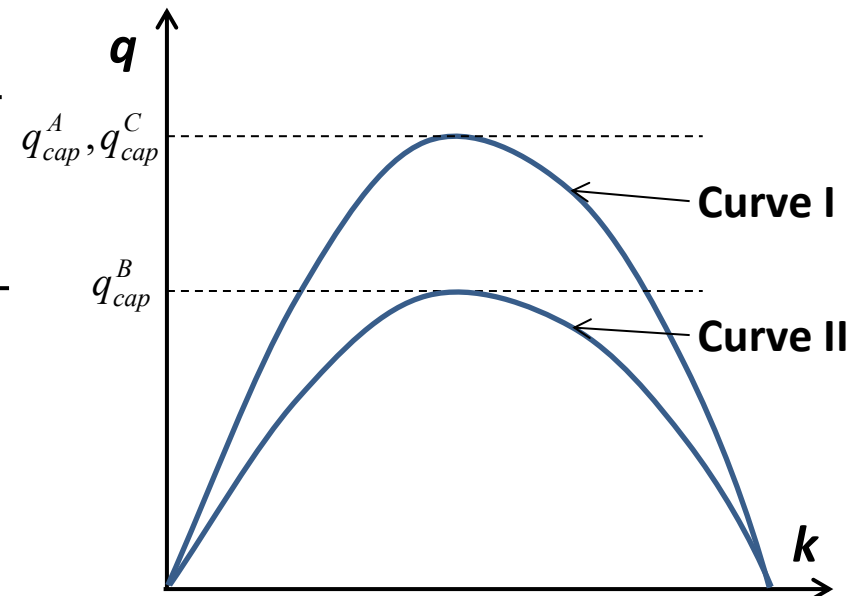
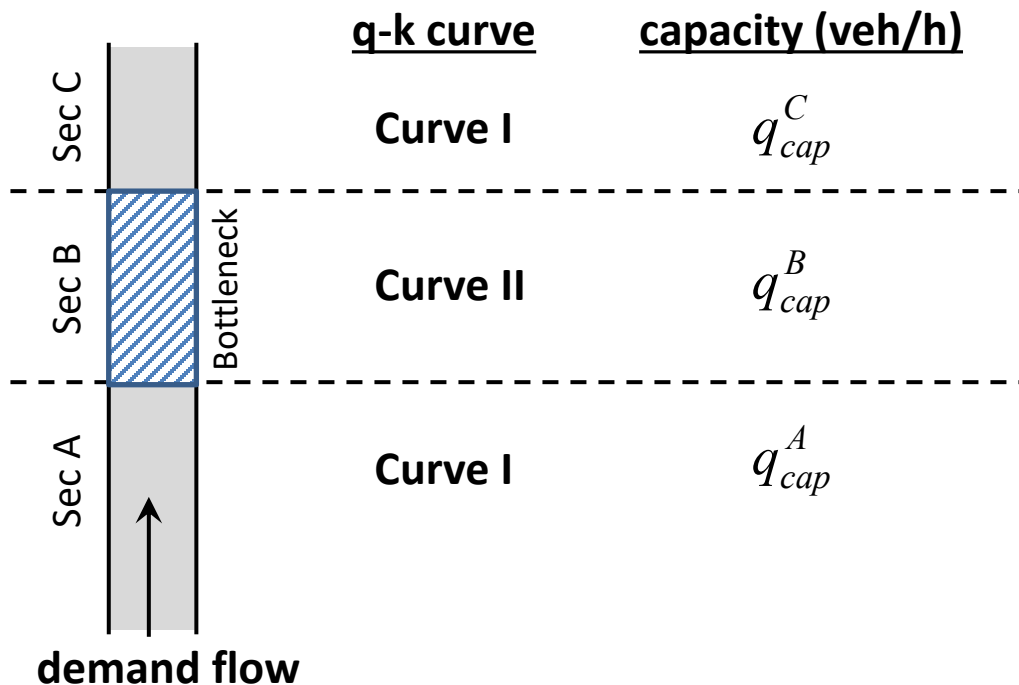


Stationary Bottlenecks

Shockwaves at Stationary Bottleneck

■ Different Flow-Density Curves observed near Classical Flow-Conserving Bottleneck

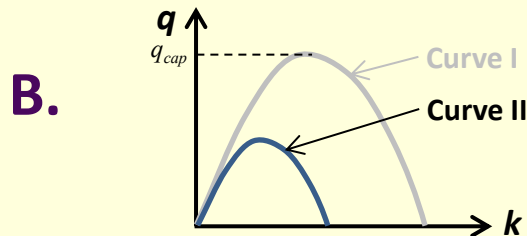
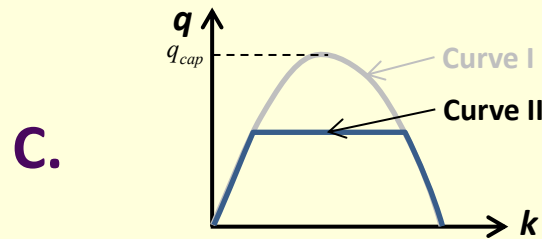
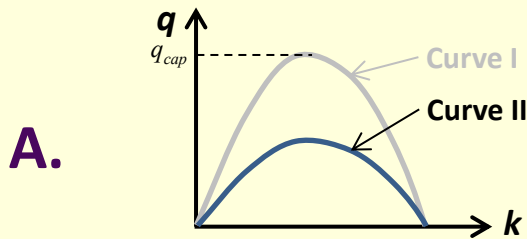
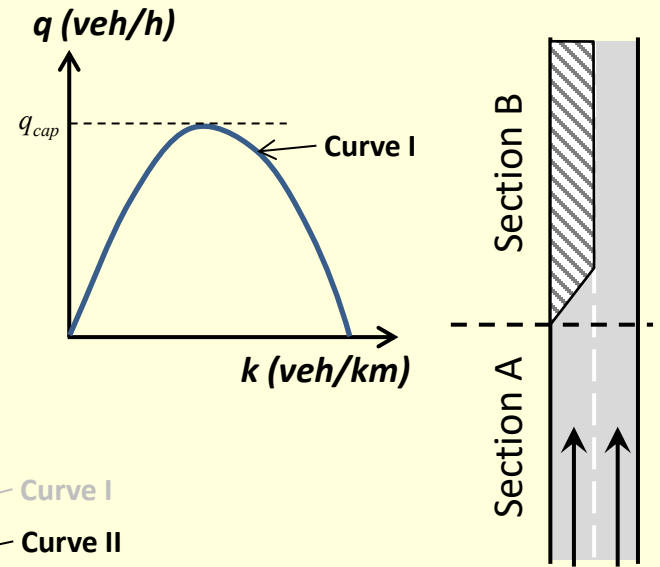
- Consider road sections A, B and C, where Section B is a classical flow-conserving bottleneck that has a lower capacity than sections A and C.
- Two flow-density diagrams exist: Curve I defining Sections A and C, and Curve II defining Section B.



Question

q-k Curves at Lane-Drop Bottleneck

A two-lane highway has the flow-density diagram defined by Curve I. If one lane is closed in Section B, what would be the q-k curve like for the bottleneck section B (Curve II)?



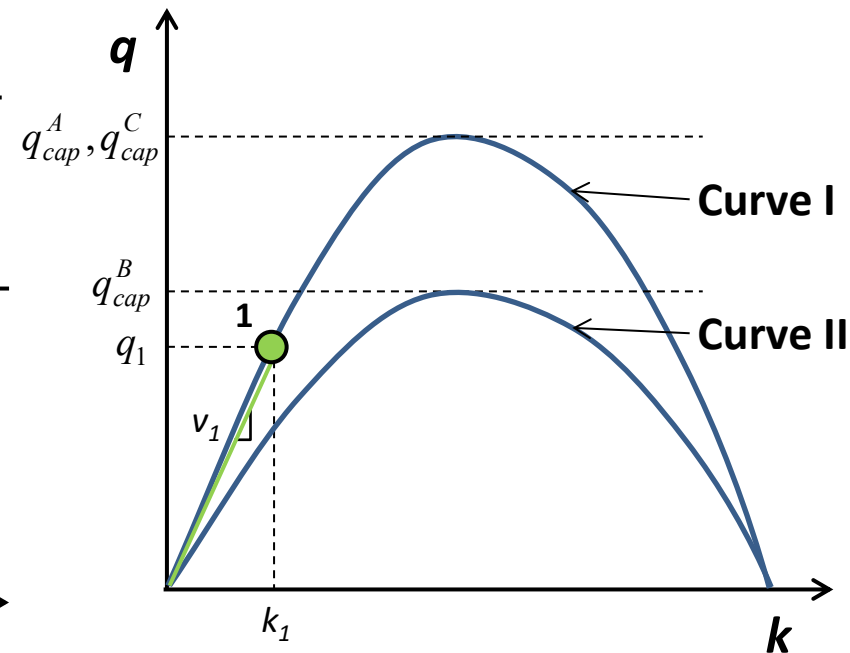
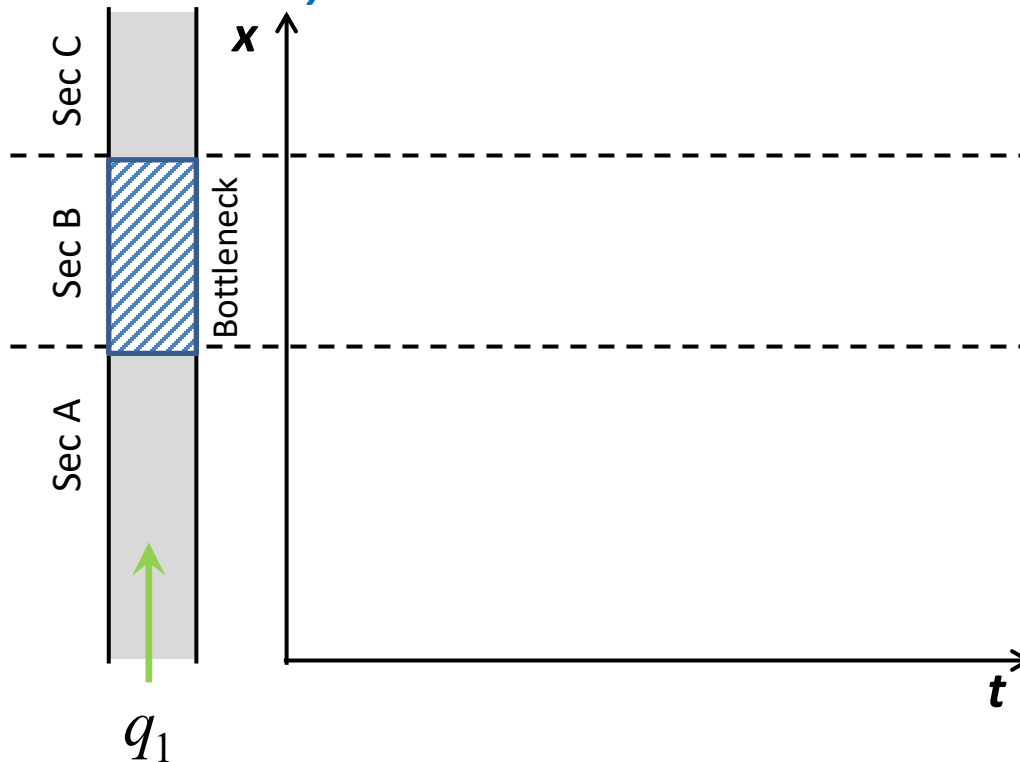
D. None of the Above

Shockwaves at Stationary Bottleneck

- The demand flow rate q_1 is less than the lower capacity: $q_1 < q_{cap}^B$

- Sections A and C have k_1 and v_1 defined by Curve I.
- Section B has k_2 and v_2 defined by Curve II at the given demand q_1 .
- At each section, the demand is served and flows on into the next section.

– *Stationary shockwaves.*

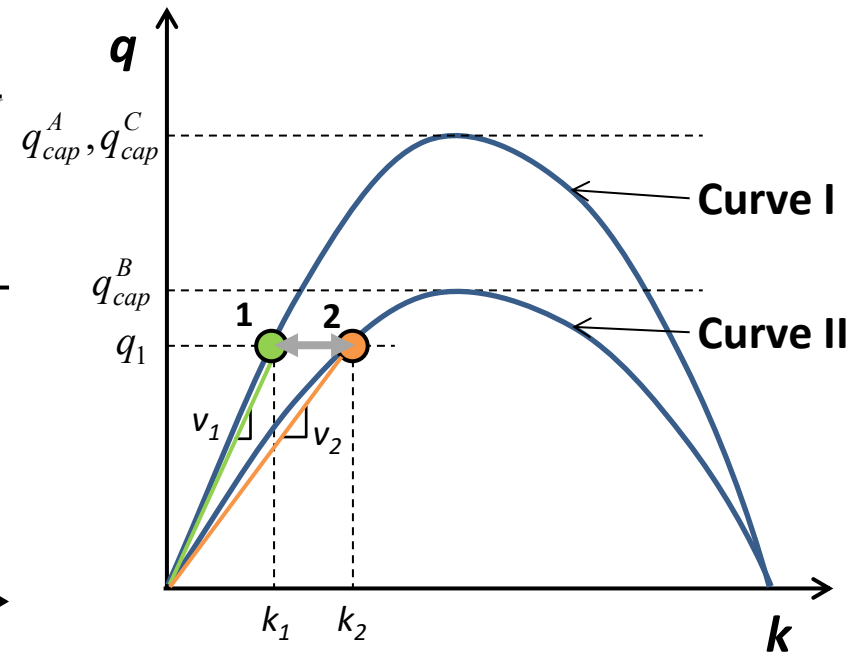
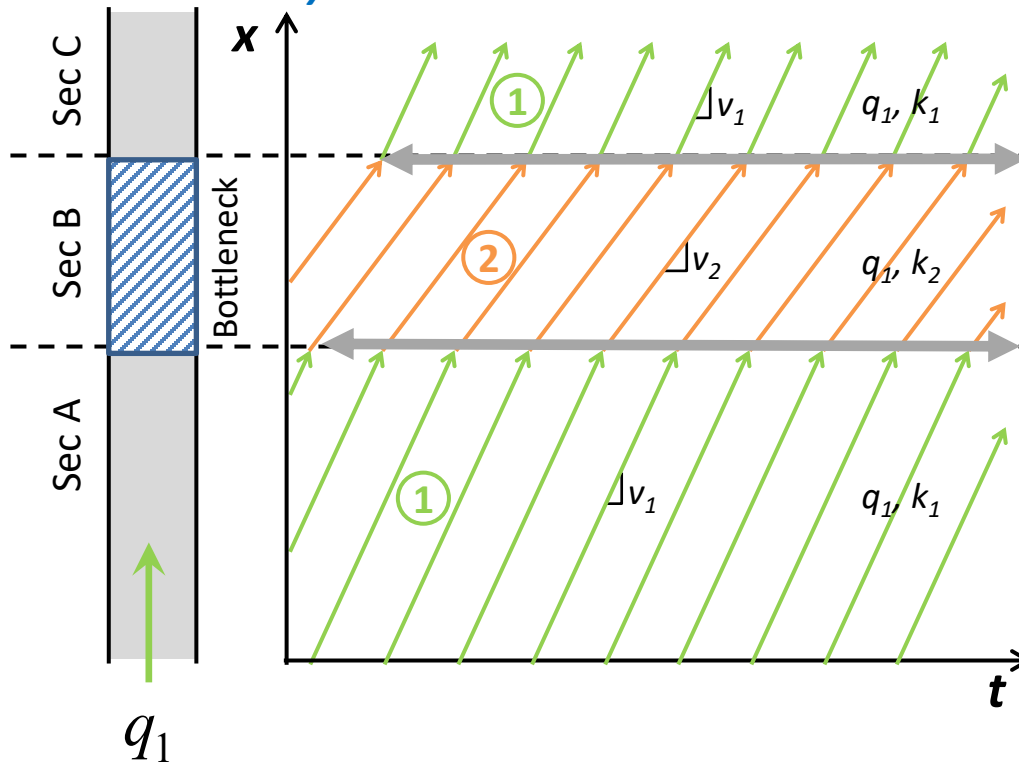


Shockwaves at Stationary Bottleneck

- The demand flow rate q_1 is less than the lower capacity: $q_1 < q_{cap}^B$

- Sections A and C have k_1 and v_1 defined by Curve I.
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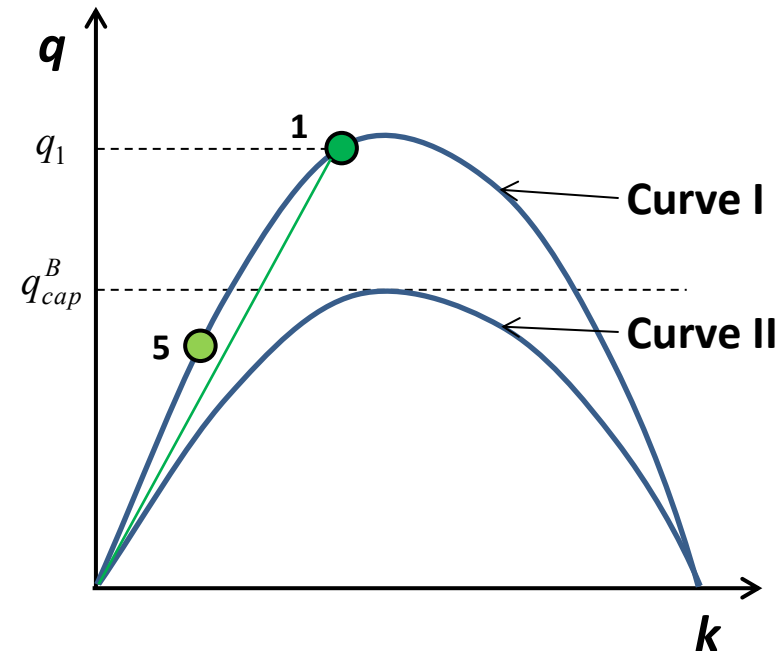
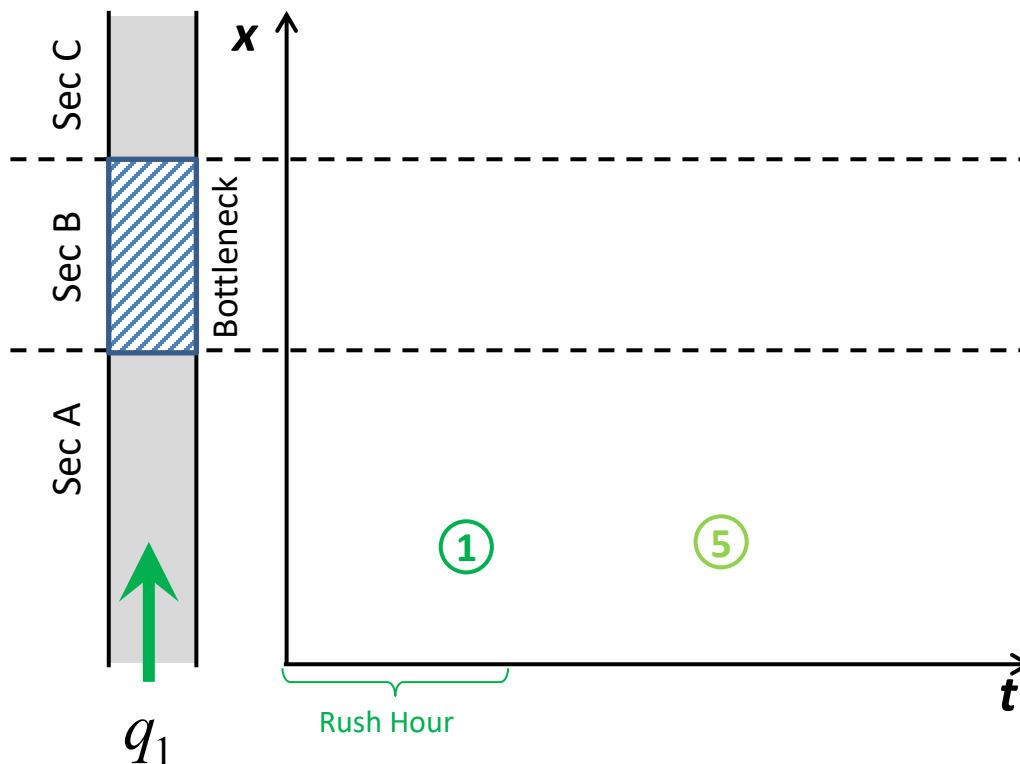
– Stationary shockwaves.



Shockwaves at Stationary Bottleneck

■ **The demand flow rate q_1 is greater than Sec B's capacity:** $q_1 > q_{cap}^B$

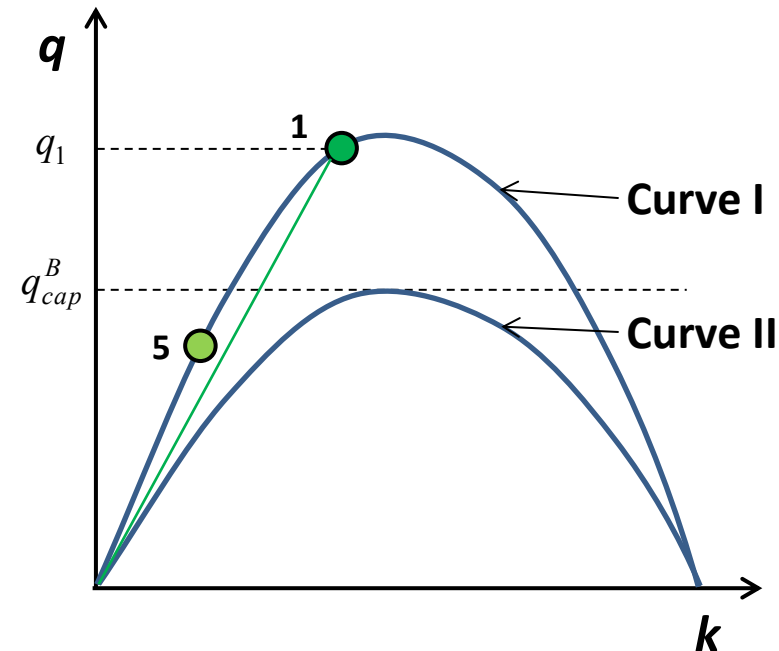
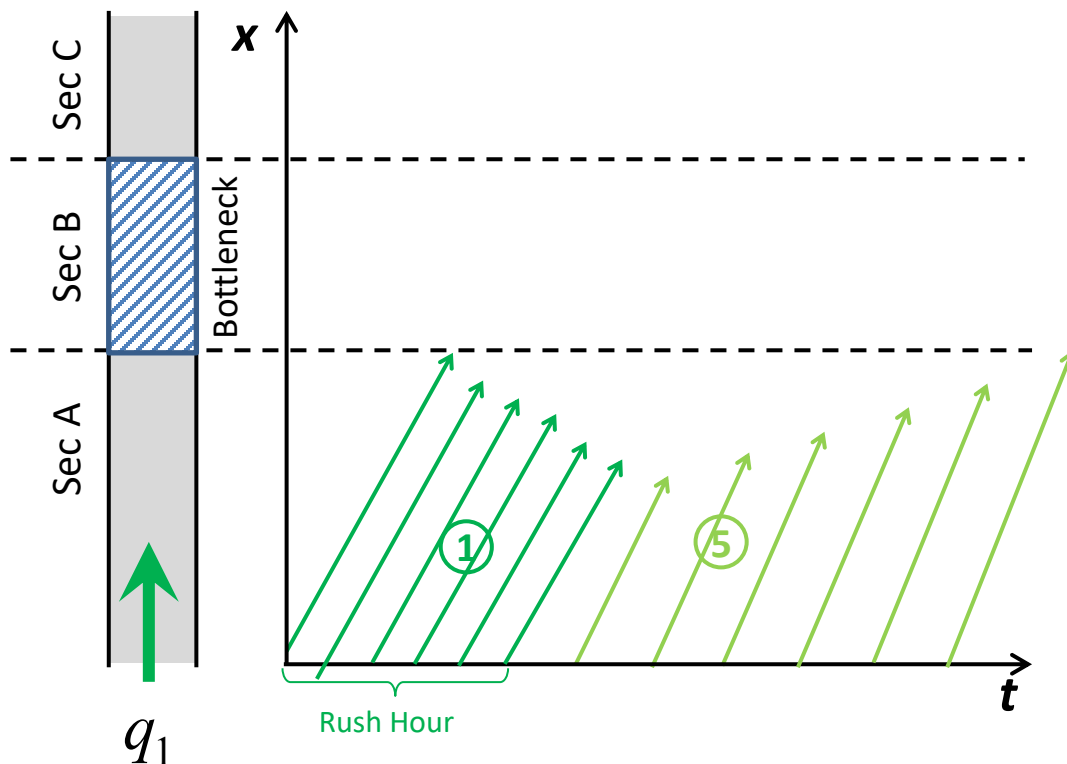
- The demand flow rate q_1 can be served at some point in Sec A operating at **Point 1** in Curve I.
- The demand q_1 cannot be passed to Sec B; the most that can be passed is $q_2 = q_{cap}^B$.
- The difference $q_1 - q_2$ is stored upstream of Sec B; Sec A operates at **Point 5** on Curve I.
- In Sec A, a jump from 5 to 1 on Curve I creates **a shockwave traveling upstream at velocity w_{12}** .



Shockwaves at Stationary Bottleneck

■ **The demand flow rate q_1 is greater than Sec B's capacity:** $q_1 > q_{cap}^B$

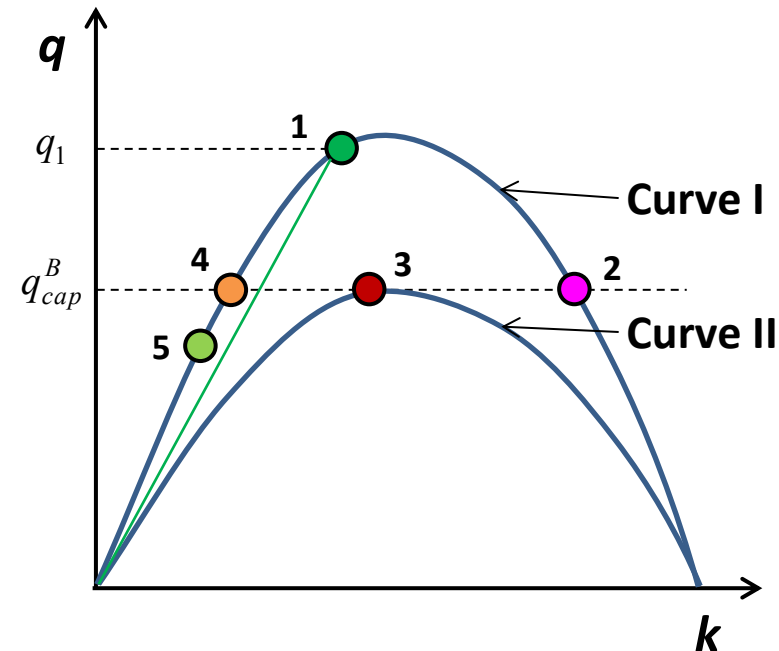
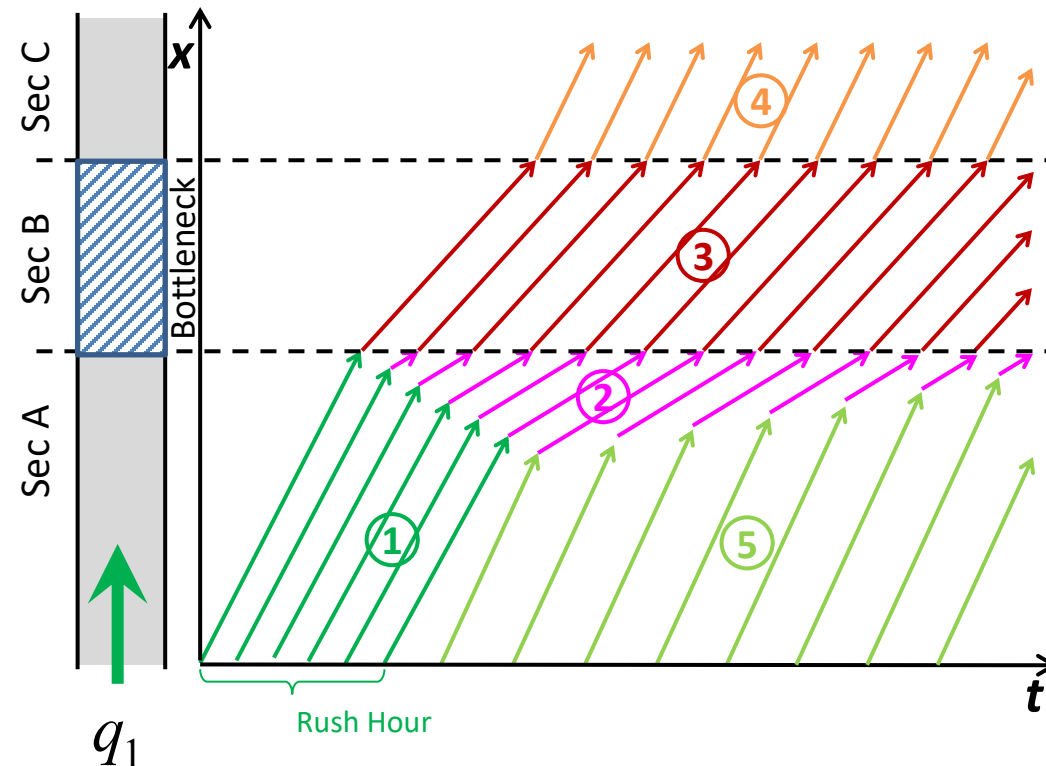
- The demand flow rate q_1 can be served at some point in Sec A operating at **Point 1** in Curve I.
- The demand q_1 cannot be passed to Sec B; the most that can be passed is $q_2 = q_{cap}^B$.
- The difference $q_1 - q_2$ is stored upstream of Sec B; Sec A operates at **Point 2** on Curve I.
- In Sec A, a jump from 2 to 1 on Curve I creates **a shockwave traveling upstream at velocity w_{12}** .



Shockwaves at Stationary Bottleneck

- The demand flow rate q_1 is greater than Sec B's capacity: $q_1 > q_{cap}^B$

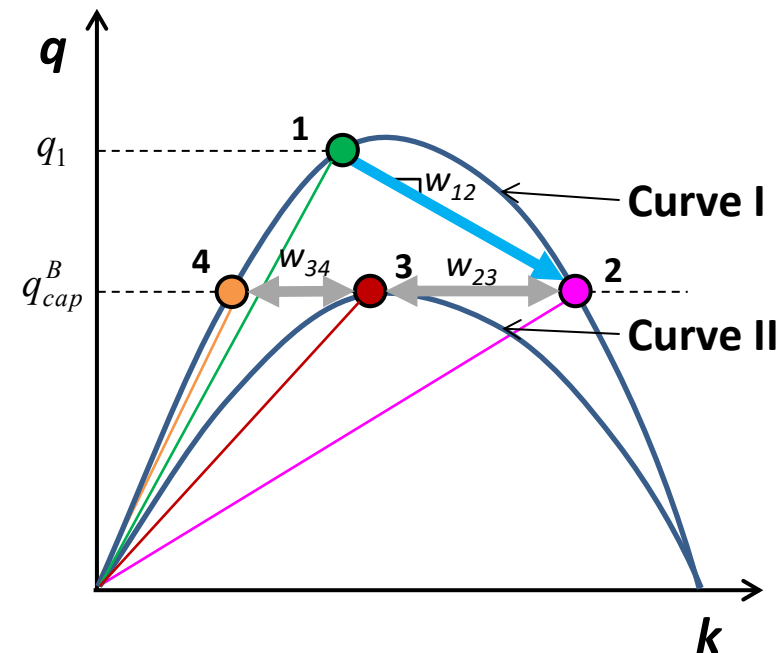
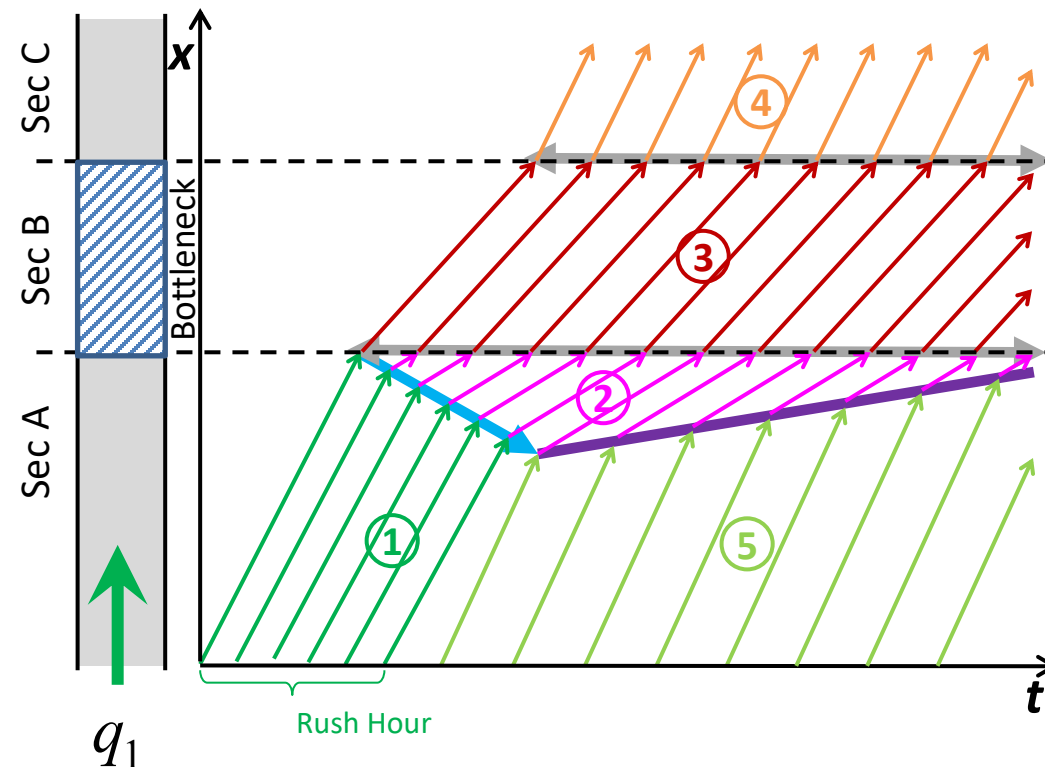
- The demand flow rate q_1 can be served at some point in Sec A operating at **Point 1** in Curve I.
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- The difference $q_1 - q_2$ is stored upstream of Sec B; Sec A operates at **Point 2** on Curve I.
- In Sec A, a jump from 2 to 1 on Curve I creates a **shockwave traveling upstream at velocity w_{12}** .



Shockwaves at Stationary Bottleneck

▪ The demand flow rate q_1 is greater than Sec B's capacity: $q_1 > q_{cap}^B$

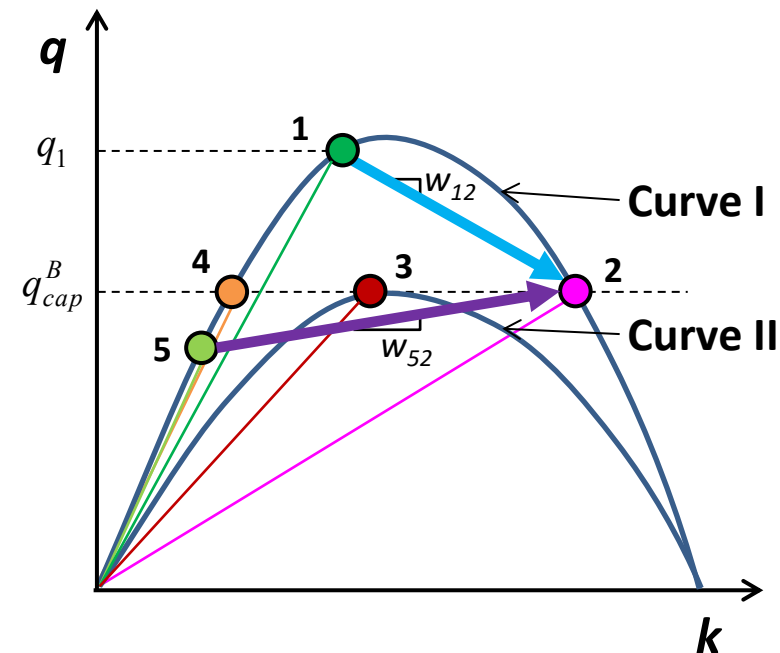
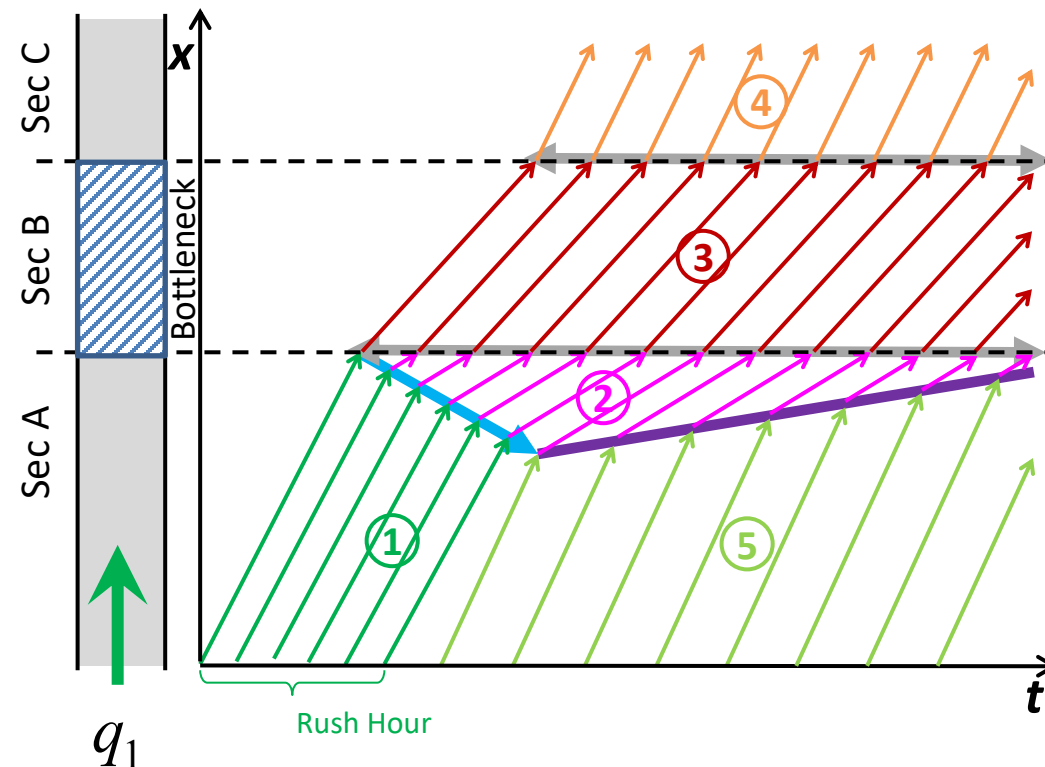
- In Sec B, the maximum flow (q_{cap}^B) is served and the section operates at its capacity (**Point 3** on Curve II).
- Only a flow level q_2 can be passed to Sec C, which operates at **Point 4** on Curve I.



Shockwaves at Stationary Bottleneck

- The demand flow rate q_1 is greater than Sec B's capacity: $q_1 > q_{cap}^B$

- In Sec A, the queue under condition 2 grows until the demand decreases to $q_5 < q_{cap}^B$. Sec A now operates at **Point 5** on Curve I.
- When the new traffic stream under condition 5 meets condition 2, a **shockwave is formed and propagates downstream at velocity w_{52}** , decreasing the size of queue.



Moving Bottlenecks

Shockwaves at Moving Bottleneck

■ Moving Bottleneck

- A **slow-moving obstruction** on a traffic stream; a slowly moving vehicle that represents a bottleneck on a road.
- Examples
 - A tractor or heavy trucks with a speed of 10-20 km/h on a two-lane road.
 - “**rolling roadblocks**” or “**traffic pacing**”: a method of temporary traffic control; a police cruiser travelling at a reduced speed (while keeping traffic blocked behind it) to pace the traffic.



[source: wikipedia]

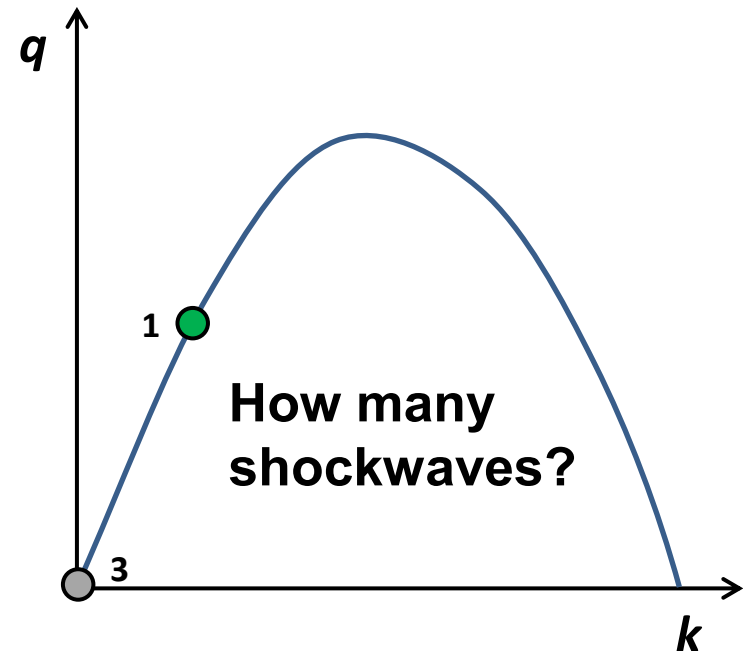
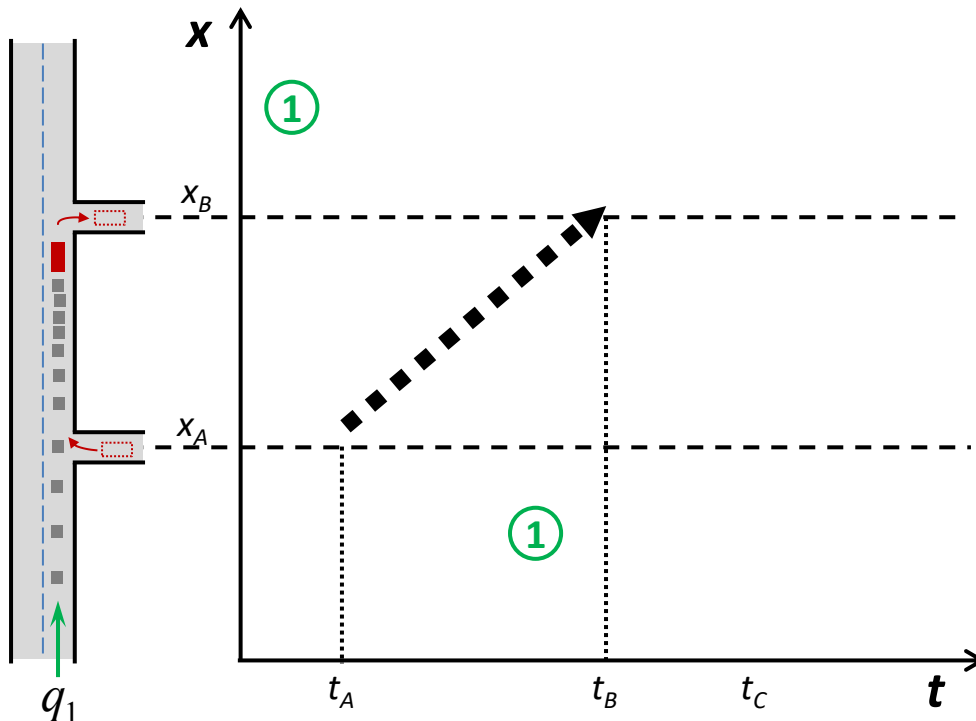


[source: https://www.flickr.com/photos/inverness_trucker/8466083182/]

Shockwaves at Moving Bottleneck

■ Slow-moving Truck Example:

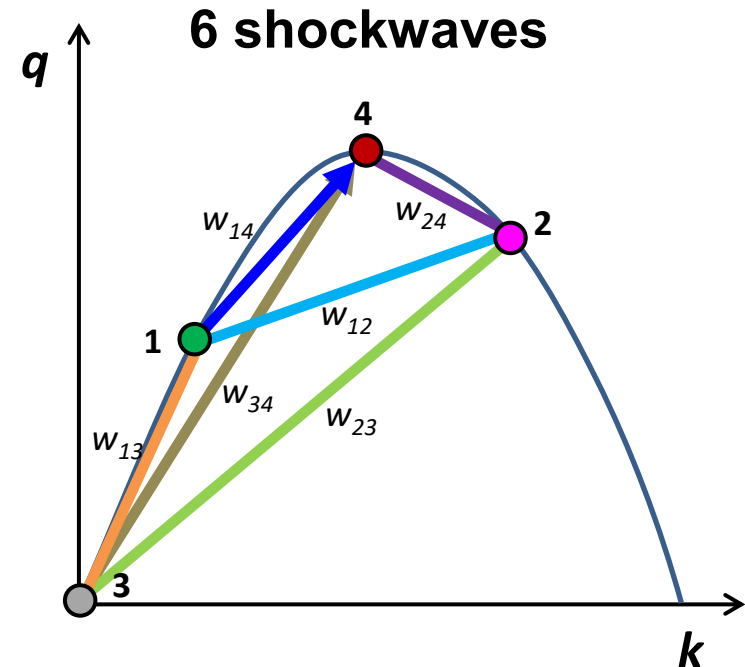
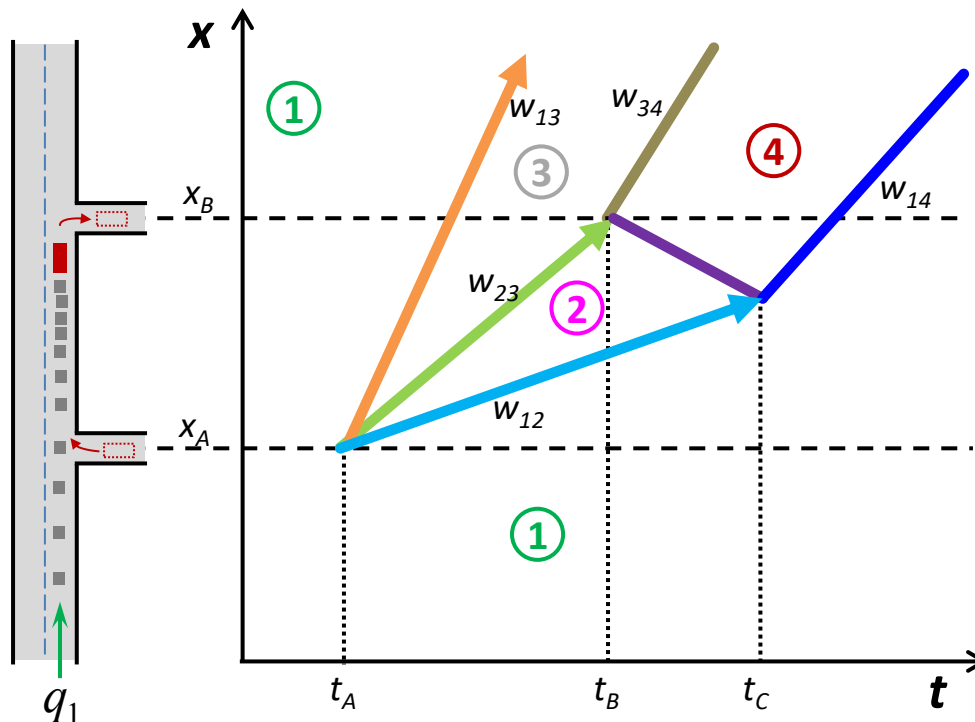
- At time t_A , a truck joins a traffic stream operating at q_1 , k_1 , and v_1 and travels at a much lower speed v_2 until it leaves the road at t_B . Vehicles cannot pass the truck and form a queue behind the truck.



Shockwaves at Moving Bottleneck

■ At time t_A , the truck enters the road:

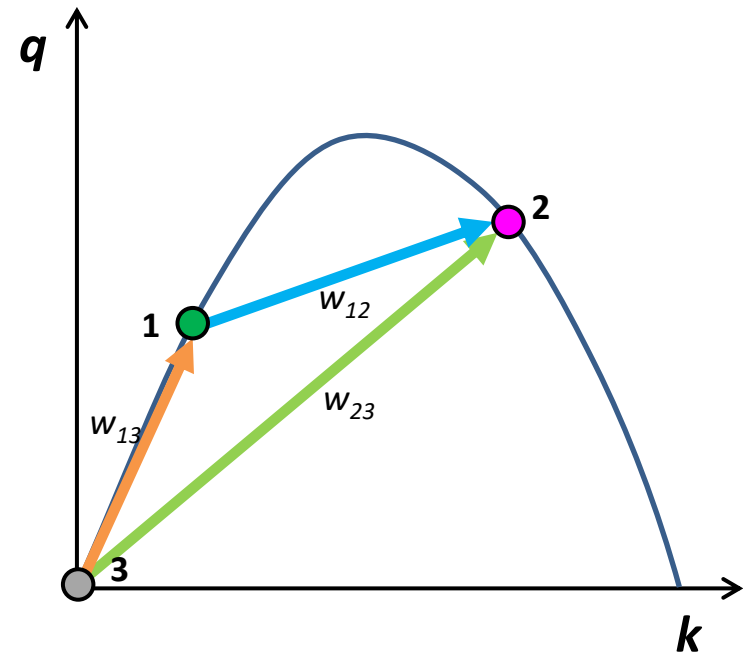
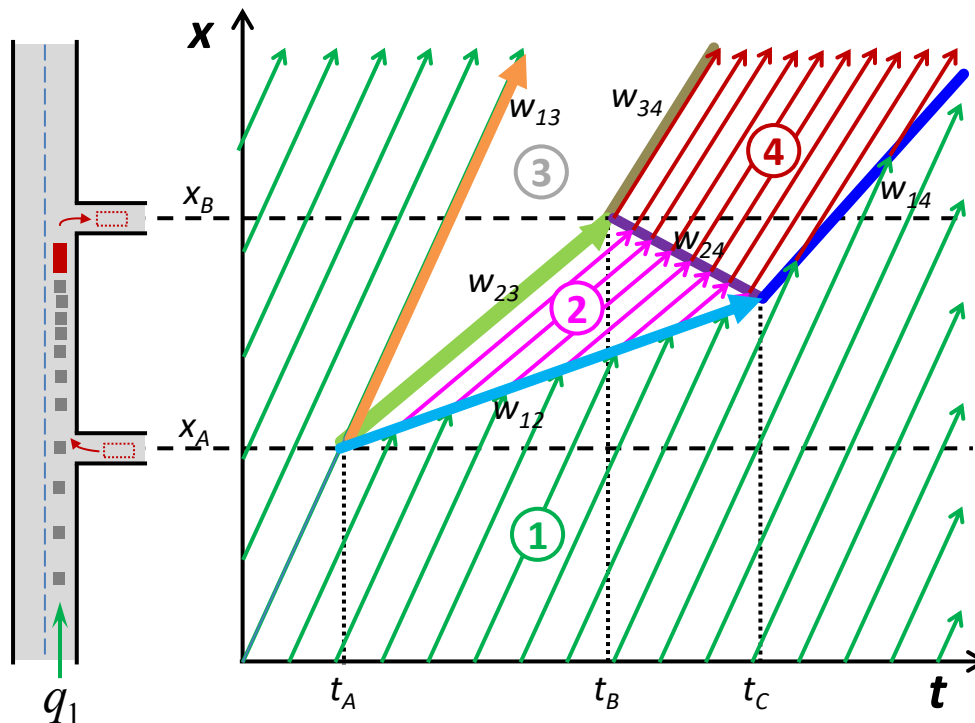
- Vehicles behind the truck slow down to v_2 + upstream traffic in 1 → **shockwave w_{12}**
- Vehicles ahead of the truck continue to travel at v_1 + no vehicle in 3 → **shockwave w_{13}**
- The truck and platoon moving under 2 + no vehicle in 3 → **shockwave w_{23}**



Shockwaves at Moving Bottleneck

■ At time t_A , the truck enters the road:

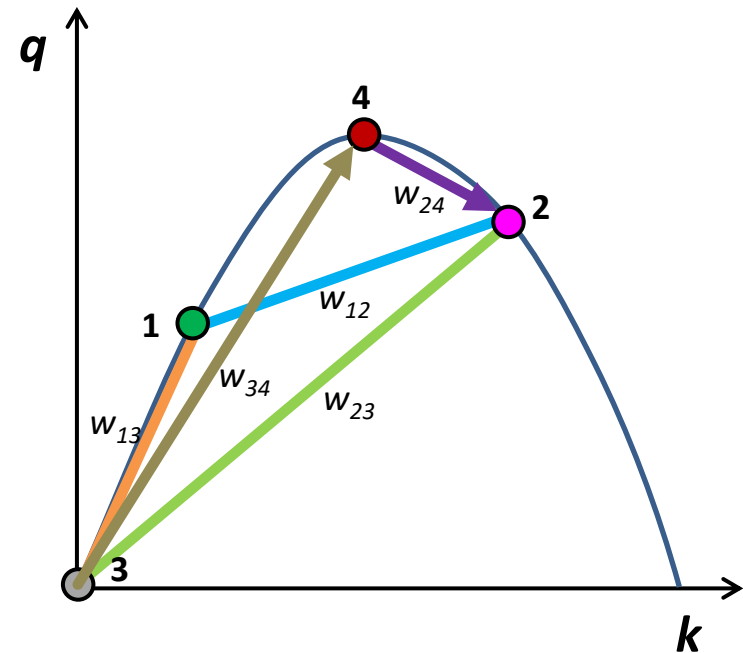
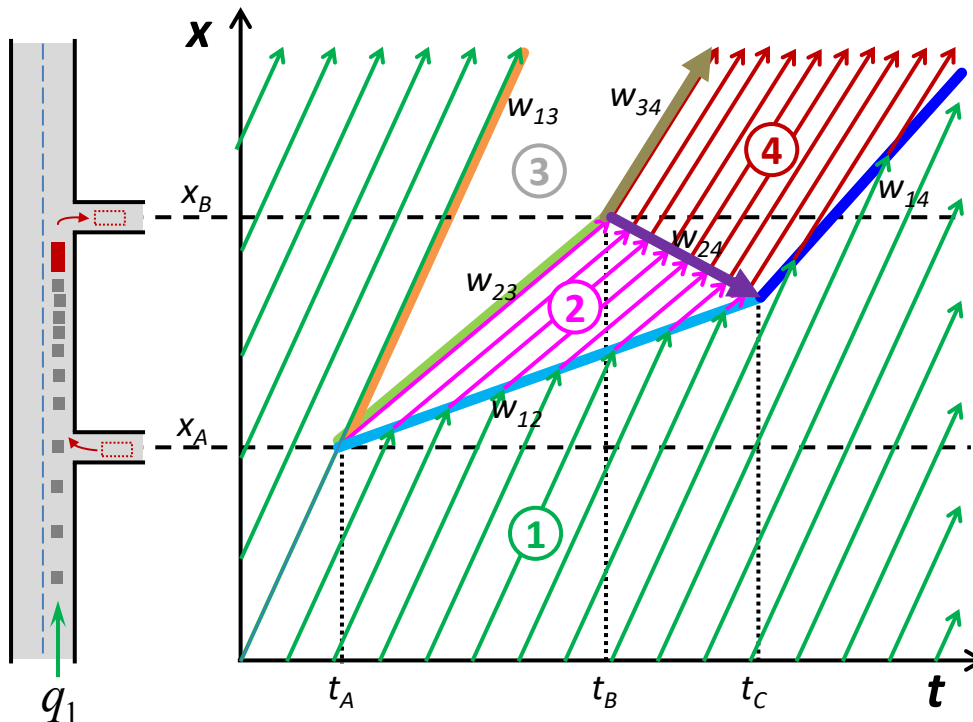
- Vehicles behind the truck slow down to v_2 + upstream traffic in 1 → **shockwave w_{12}**
- Vehicles ahead of the truck continue to travel at v_1 + no vehicle in 3 → **shockwave w_{13}**
- The truck and platoon moving under 2 + no vehicle in 3 → **shockwave w_{23}**



Shockwaves at Moving Bottleneck

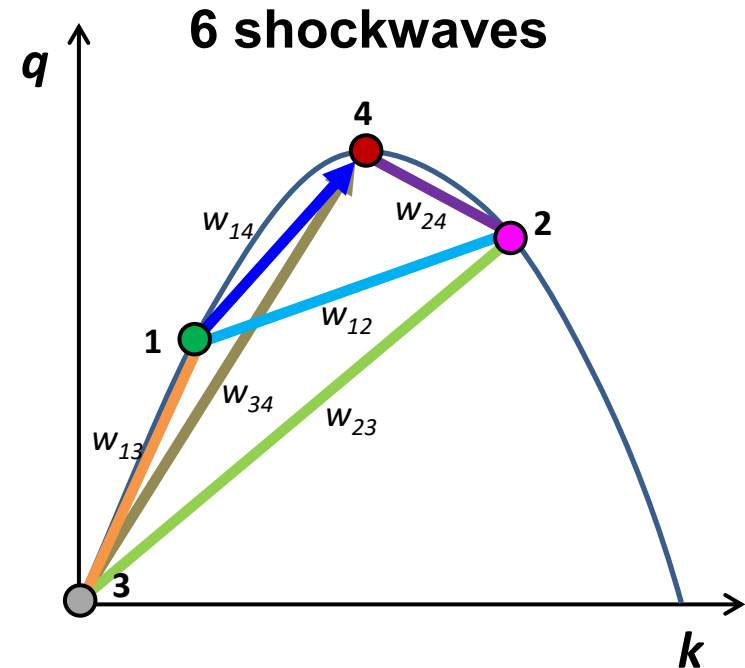
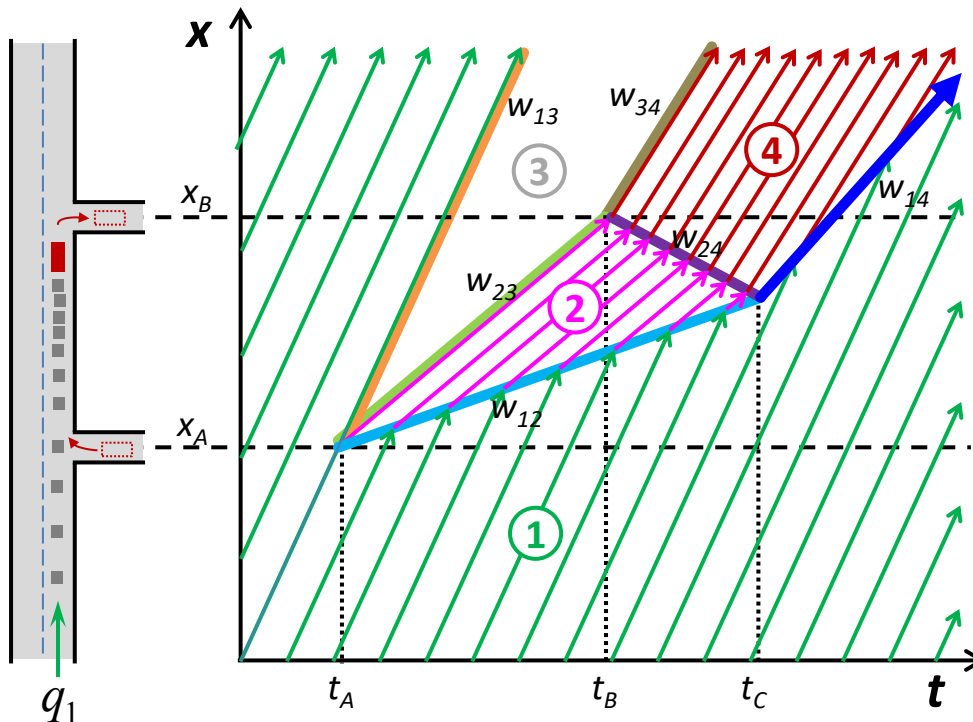
■ At time t_B , the truck leaves the road:

- The flow will be increased to the capacity of the road with condition 4
 - creates a forward moving shockwave w_{34}
 - creates a backward moving shockwave w_{24}



Shockwaves at Moving Bottleneck

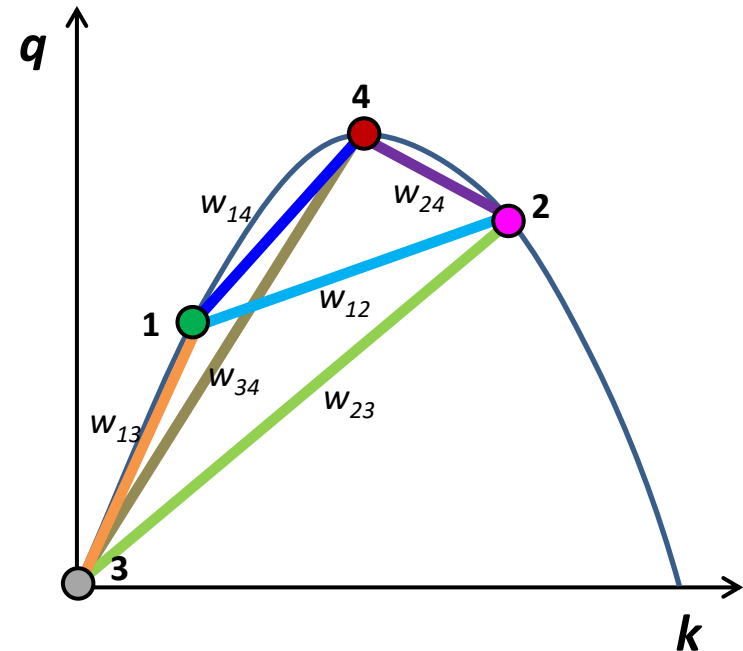
- At time t_c , two shockwaves w_{12} and w_{24} coincide:
 - The queued condition 2 ends.
 - A new forward moving shockwave w_{14} is formed.



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Platoon: density $k_2 = 100$ veh/km

- (a) What is the rate at which the platoon grows, in units of vehicles per hour?
- (b) How many vehicles will be in the platoon by the time the truck leaves the road?



Exercise

Original traffic condition: $q_1 = 1500$ veh/h, $k_1 = 25$ veh/km,

Truck: speed (v_2) = 20 km/h, travel distance $L = 2.5$ km

Platoon: density $k_2 = 100$ veh/km

(a) What is the rate at which the platoon grows, in units of vehicles per hour?

(b) How many vehicles will be in the platoon by the time the truck leaves the road?

[Solution to (a)]

The velocity of a shockwave upstream of the platoon is:

$$w_{12} = \frac{q_2 - q_1}{k_2 - k_1} = \frac{(100 \times 20) - 1500}{100 - 25} = \frac{500}{75} = 6.67 \text{ km/h}$$

Since the platoon is moving at 20 km/h and the shock front moves forward relative to the road at 6.67 km/h, the platoon growth speed is: $20 - (6.67) = 13.33$ km/h

The rate at which the queue grows, in units of veh/h is:

$$\text{platoon density} * \text{growth speed} = 100 * 13.33 = \boxed{1333 \text{ veh/h}}$$

$$(\text{or } q_1 - k_1 w_{12} = q_2 - k_2 w_{12} = 1333 \text{ veh/h})$$

Exercise

Original traffic condition: $q_1 = 1500$ veh/h, $k_1 = 25$ veh/km,

Truck: speed (v_2) = 20 km/h, travel distance $L = 2.5$ km

Platoon: density $k_2 = 100$ veh/km

(a) What is the rate at which the platoon grows, in units of vehicles per hour?

(b) How many vehicles will be in the platoon by the time the truck leaves the road?

[Solution to (b)]

The time spent by the truck on the road = $2.5[\text{km}] / 20[\text{km/h}] = 0.125$ h

The number of vehicles in platoon after 0.125 h is

= the platoon growth rate * 0.125 = $1333 \text{ veh/h} * 0.125 \text{ h} = \mathbf{167 \text{ veh}}$

Next

This week

- Tutorial – Shockwaves